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1.0 Introduction

In the current digital economy, e-commerce systems produce huge amounts of transactional data in a day. When properly processed and analyzed, this data can give some valuable information about customer behavior, the performance of the product and the conduct of the business as a whole. Nonetheless, conventional data processing systems are usually not able to process such data due to its volume, speed, and complexity. Consequently, the use of big data technologies has been on the increase in organizations to facilitate scalable, efficient and real-time analytics.

This report includes a complete big data architecture design of NovaRetail Analytics Sdn. Bhd. which is a e-commerce platform with a large scale operating in Malaysia. The company focuses on online shopping services in the subscription of different products and has a wide range of customers. As its operations increase fast, NovaRetail is experiencing difficulties in managing and deriving useful information out of the growing amount of data, especially when it comes to the analysis of the overall sales performance of various product lines.

This report provides a strong suggestion for a big data analytics system based on the use of Hadoop, HDFS, MapReduce, and MongoDB tools to help solve the challenges. The architecture will allow high-scale data ingestion, distributed storage, efficient data processing, and scalable data retrieval. One analytical element of the system is total sales per product, which gives critical information to the business decision-making, inventory management, and optimization of revenues.

In addition, the report provides the business needs, system architecture, data flow processes, technology justification, implementation plans and the challenges which could be encountered with the proposed solution. The proposed architecture will contribute to the improvement of data-driven decision-making and increase operational efficiency in the organization by incorporating all of these elements.

2.0 Business Requirements Document

2.1 Company Profile

NovaRetail Analytics Sdn. Bhd. e-commerce analytics company based in Malaysia with a massive online store platform. The company has a Business to Consumer (B2C) model which links customers to a broad variety of products in different categories which are home decor, lifestyle products and seasonal products. Its business purpose is to use data-driven information to improve customer experience, provide optimization of products, and optimize the performance of revenue.

The company receives thousands of transactions every day of customers in various regions. As online shopping has become more popular, NovaRetail has grown at a very high rate in terms of the number of people using the application and the amount of data generated. Such an increase requires the provision of a reliable and scalable data processing system that can process large volumes of structured data on transactions.

2.2 Current Data Challenges

Although it has grown, NovaRetail has a number of issues with the available data processing systems. Conventional database systems cannot effectively analyze large data streams on time resulting in delays in the production of analytical information. This weakness is a debilitating factor to the company in terms of timely decision making in business. Also, currently, data is held in disjointed forms in various systems leading to discrepancies and problems in integrating data.

The absence of a centralized data structure also makes it hard to carry out high volume computations, like the total sales per product of millions of records. Moreover, system performance decreases and processing times increase tremendously with increase in the data. This makes the analytics workflows to have bottlenecks and reduces scalability of the company.

2.3 Data Volume and Characteristics

NovaRetail creates large quantities of transactional data in a daily manner. The platform records about 50,000 to 100,000 transactions daily and this is subject to increase with time to millions of records. The data is mostly structured, such as transaction IDs, customer IDs, product description, prices and timestamps.

The data will be gathered through a wide range of platforms, such as web applications, mobile platforms, and back-end transaction systems. The data is collected over time into massive datasets applicable in big data processing systems like Hadoop.

These properties of the data can be characterized with the help of the 3Vs of big data:

- Volume: Huge volumes of transactional data are usually produced on a continuous basis.
- Velocity: Fast generation of data based on real time user interactions.
- Variety: Multiple-attribute and format structured data.

2.4 Key Analytics Question

NovaRetail wants to derive valuable insights out of data in order to stay competitive. The key analytical goal is to calculate the overall sales per product category that will assist in showing the high-performance products and revenue drivers.

Among that, the company aims to provide the answers to some of the crucial business questions:

- What are the most revenue-prone products?

- What are the sales trends during the time period?

- What category forms the major percentage of business performance?

- What can be done to optimize inventory with regard to sales patterns?

Such insights can help in improved strategic planning, marketing decisions and inventory management.

2.5 Performance Requirements

The big data system proposed should have the capacity to process data that are large in size effectively and still reliable and scalable. The architecture is built in such a manner that it can support the batch processing with the aid of Hadoop MapReduce, that is, the system can process thousands of records in a workable time frame, which is several in the form of millions.

The structure will consist of a Hadoop cluster comprising of about five to eight nodes, which will be able to handle enough computing power to do distributed processing. The storage system will be able to accommodate approximately 5 terabytes of data at the beginning and store up to 20 terabytes of data as the volume of data expands. In order to provide data reliability and fault tolerance, a three-factor replication will be applied in HDFS.

Besides, the system has to be highly available, with a goal of achieving a minimum of 99.5% availability. It should also be able to maintain high throughput in processing tasks especially in situations where it is required to handle peak data loads and also in cases where the data is required to provide analytical outputs that are not accompanied by long delays.

2.6 Data Retention and Compliance

NovaRetail has the Personal Data Protection Act (PDPA) of Malaysia as it is obliged to take care of customer data and make sure that they are handled in a secure and responsible manner. The access control mechanisms and policies of data governance offer protection to sensitive data, including customer identifiers.

The company has a data retention policy that is three to five years and gives it an opportunity to do a complete history analysis and prediction. Raw transactional data are stored as HDFS to provide large scale batch processing and processed and aggregated data is stored in MongoDB to provide quicker access and retrieval. Data anonymization methods are used where applicable to maintain user privacy and to ascertain that regulatory standards are met.

2.7 Budget Constraints

NovaRetail uses an open-source technology stack to ensure low cost, with Hadoop, HDFS and MongoDB being used to facilitate high cost rather than high license commercial systems. The major investment is on hardware infrastructure and system maintenance.

The overall preliminary setup will cost between RM 60,000 and RM 120,000 which will include the computing nodes, storage systems and the networking parts. Other Annual expenses, such as the operational and maintenance expenses will exclude RM 15,000 to RM 25,000. The company will use a scalable infrastructure model in order to support its long-term growth, whereby resources will be added as data volume and demand on processing resources grows.

3.0 Architecture Design

The proposed big data architecture of NovaRetail Analytics Sdn. Bhd. is a multi-layered system, which will allow the efficient data ingestion, scalable storage and processing of the data as well as providing flexibility in the way data is accessed. The architecture combines Hadoop, HDFS, MapReduce and MongoDB to facilitate the large-scale analytics as well as providing reliability and performance. All the architecture layers are important in converting raw transactional data into valuable business information.

3.1 Data Ingestion Layer

Data ingestion layer is the layer that has to get data as collected by various sources and move it into the big data ecosystem. In NovaRetail, transaction data is based on different platforms such as web-based e-commerce systems, mobile applications, and systems to process transactions on the back-end.

The data is mostly loaded in a batch mode, where a record of the transactions is regularly moved out in CSV format and loaded into the Hadoop Distributed File System (HDFS). Such a strategy will be appropriate when working with massive amounts of organized information. Basic data validation and cleaning operations are done before ingestion to make sure there is consistency in the data such as the elimination of missing values and different formatting errors.

This layer forms a basis to prevent any dirty and unstructured data from accessing the system and avoid any error in data processing at a later stage.

3.2 Storage Layer

Storage layer is developed on the Hadoop distributed file system (HDFS) and this offers a fault-tolerant and scalable storage facility of large data. HDFS divides data into more than one node within the cluster and this allows the data to be accessed simultaneously and enhances the performance of the system.

A three-replication factor is used to make it reliable that is, each data block is replicated in three nodes. This ensures that the system does not lose data in case of failure of the hardware. The system is programmed, which has a starting storage capacity of about 5 terabytes, but it can expand with an increase in data volume.

Besides HDFS, MongoDB is also deployed as an alternative storage system of processed and structured data. HDFS is more suited to large-scale batch processing, whereas MongoDB is more suited to fast data retrieval and functioning under flexible queries. This renders MongoDB to be appropriate in aggregating results, metadata, and analytics outputs which are needed to be accessed quickly.

3.3 Processing Layer

The processing layer makes use of Hadoop MapReduce framework in solving the analysis of large-scale data. MapReduce provides distributed computing by breaking down tasks into little sub-tasks which can be computed in parallel on various nodes on Hadoop cluster.

A MapReduce job in this project is planned to compute the sum of the sales of each category of products. At the mapping stage, all the records of transactions are handled to retrieve the product type and the price. The reducing phase will entail the system adding up all the sales of each product category and giving an output in a summarized form.

This methodology will allow dealing with large datasets efficiently since the calculations will be spread to the cluster. It can manipulate records of millions of records in a very brief duration and, hence, can be used in batch analytics.

3.4 Analytics and Visualization Layer

The analytics and visualization layer performs the duty of transforming processed information into useful information to the business users. The total sales per product category is the main key performance indicator (KPI) in this system that can give useful information on the performance of the products and their contribution to revenue.

MongoDB is a storage of processed data, which can be easily queried and integrated with visualization software like Power BI or Tableau. This allows users to create dashboards, reports and visualization of sales data, which is useful in decision-making.

Though visualization tools are not applied to this project directly, the architecture is implemented in such a manner that they will be easily integrated with the visualization platforms so that they can be scaled to accommodate future improvements.

4.0 Data Flow Diagrams

The data flow diagrams will be vital in showing data flow in the proposed big data architecture. They give a vivid visual illustration of the interaction of various components of the system, starting with the ingestion of data and ending with the ultimate analytics information. This project uses two important diagrams to illustrate the end-to-end data flow as well as internal process workflow in the Hadoop ecosystem.

4.1 End-to-End Data Flow Diagram

The end-to-end data flow diagram is used to determine the way raw transactional data is gathered, processed, and converted to meaningful insights. The first part of the process involves the creation of data by various services, such as web applications, mobile platforms, and transaction systems at the backend. These data sources are constantly producing transactional records as the customers are interacting with the e-commerce site.

The data collected is then ready and fed into the system in batch. The dataset is uploaded to the HDFS after a series of basic data cleaning and validation steps are performed and it is distributed in a distributed form across the multiple nodes. This is to provide scalability and fault tolerance.

After the data is stored in HDFS, MapReduce framework is used to process the data to calculate the total sales of each category of products. This processed output is then transferred to MongoDB where it is stored in a structured format to be able to query and retrieve it effectively. Lastly, the data is also made accessible to visualization and reporting, which allows business users to derive insights and assist in making decisions.

4.2 MapReduce Workflow Diagram

The MapReduce workflow diagram is concerned about the internal data processing process of the Hadoop ecosystem. It emphasizes the partitioning, processing and aggregation of data in the cluster.

First, the input data in HDFS are divided into relatively small data blocks. These blocks are partitioned to various nodes within the Hadoop cluster thus enabling parallel processing. In the mapping stage, the data is divided into smaller parts processed by each mapper, and essential pairs of key-values are separated, the product category is the key, and the corresponding price is the value.

After this there is the shuffle and sort stage which arranges intermediate data by sorting together all the values that have the same key. This will make sure that records of the same product category are done simultaneously. During the reducing stage, the values are summed up by the reducer to determine the overall amount of sales in each product category.

The resulting output is then saved to HDFS before it is saved to MongoDB where it is further analyzed. This workflow shows the effectiveness of distributed processing with processing large-sized datasets.

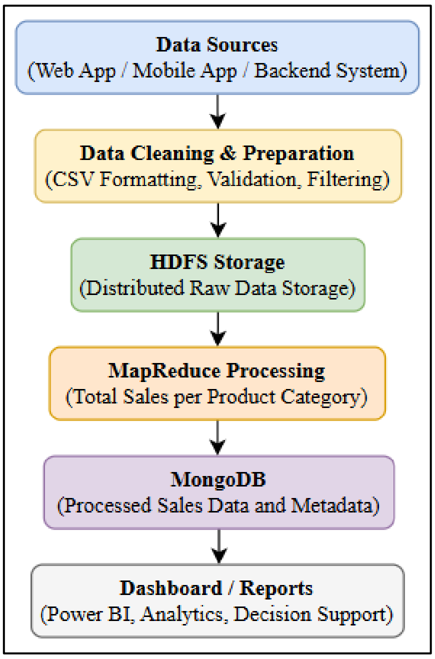


Figure 1: End-to-End Data Flow for NovaRetail Big Data Architecture

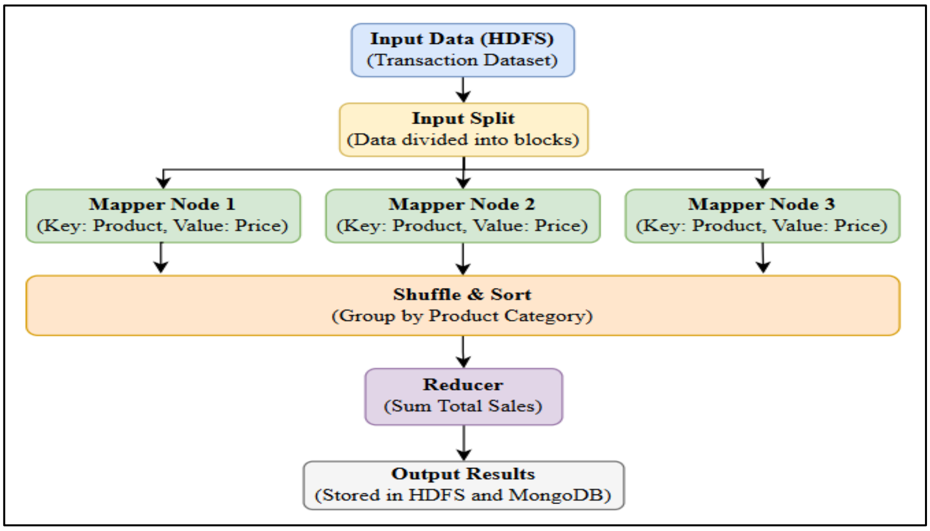


Figure 2: MapReduce Workflow for Sales Aggregation

5.0 Technology Justification

The choice of technologies used in the proposed big data architecture can be explained by the necessity to process a great amount of transactional data efficiently, provide the system with scalability, and allow the reliable processing of analytics. Hadoop, HDFS, MapReduce and MongoDB are selected as the foundational components since they complement on the distributed storage, processing and access.

5.1 Justification for Hadoop and HDFS

The choice of Hadoop as the basis of the architecture is because it can handle and store huge amounts of data within a distributed infrastructure. It offers scaled architecture which enables the distribution of data around multiple nodes and therefore provides parallel processing and better performance. This is especially crucial to NovaRetail where one can deal with a number of millions of records in a year of transactions.

As one of the main elements of Hadoop, HDFS is employed in data storage over distance. It groups big datasets and splits them into small segments and disseminates them among several nodes in the cluster. This provides scalability as well as fault tolerance. The system has a replication factor of three making it resilient to hardware failure without losing data, which is essential to preserve the data integrity in a production environment.

Hadoop and HDFS are more flexible and cost-effective in comparison with the traditional relational databases when they are used to process large volumes of data in batches. They are also open source, which minimizes the implementation cost, thus appropriate to a growing e-commerce company such as NovaRetail.

5.2 Justification for MapReduce

The main data processing model that will be used is MapReduce because the model is efficient in processing large volumes of data by firing parallel calculations. It also enables the breakup of complex analytical work into smaller sub works, which are run concurrently in a number of nodes in the Hadoop cluster.

The MapReduce technique is especially appropriate in this project to calculate total sales by each category of products. Mapping phase separates out appropriate key-value pairs based on the dataset and the reducing phase summarizes the information to give significant results. This parallel processing system saves a lot of computational time as compared to the old sequential methods of processing.

Moreover, MapReduce has inbuilt fault tolerance and data locality capabilities and as such, tasks performed by the processing functions are done near the location of the data stored. This reduces the overhead in the network and enhances the performance of the system.

5.3 Justification for MongoDB

MongoDB is incorporated in architecture to be the complement of HDFS to offer an efficient and flexible solution to the storage of processed outcomes. HDFS is not designed to support quick querying and real-time data access, although it is optimized to store data at large scale in batch.

Being a NoSQL database, MongoDB enables the storage of data in the form of a document, which results in a high degree of flexibility in the structure of the schema and accelerates the retrieval of data. This is why it is best suited to the storage of aggregated data, for instance total sales per product, which are frequently accessed to be used in reporting and visualization.

Besides, MongoDB is scalable for indexing and effective querying which increases the performance in getting analytical insights. This will enable business users to access data within a short period of time without involving complicated batch processing process on every occasion.

5.4 Technology Integration and Alternatives

Hadoop plus HDFS, MapReduce, and MongoDB integration form an effective and unified big data ecosystem. The HDFS is the main storage layer of raw data, MapReduce processes the data in a distributed way and the processed outputs are stored in MongoDB to be readily accessed. The layered approach means that every technology is applied where they are best suited and leads to a well-balanced and scalable system.

Other alternative technologies like Apache Spark, data warehouses based on clouds, and regular relational databases were also taken into consideration. Although Apache Spark is better at processing data in memory, MapReduce is selected in this project because it is simple, stable and well suited with batch processing tasks. In a similar fashion, the choice of cloud-based solutions was not determined by a possible economic limitation and the emphasis on on-premises open-source architecture.

In general, the chosen technologies offer an economical, scalable, and reliable solution that will satisfy the business and technical needs of NovaRetail.

6.0 Implementation Plan

The proposal for the big data architecture to be adopted by NovaRetail Analytics Sdn. Bhd. would be implemented in a systematic and progressive manner to achieve system stability, scalability, and effective implementation. All stages are designed in such a way that the architecture is built progressively, starting with the configuration of the infrastructure and delivering analytics and enabling testing and optimization on each stage.

6.1 Infrastructure Setup and Data Ingestion (Phase 1)

The first step is oriented to the automated setup of the Hadoop ecosystem and the pre-processing of the environment. This involves configuring a multi-node cluster of Hadoop and its related environment setup including setting NameNode and DataNodes as well as the environmental variables including HADOODPHOME and JAVAHOME. There are core configuration files, such as core-site.xml and hdfs-site.xml that are configured to define the HDFS architecture, such as replication factor and storage paths.

Ingestion of data into HDFS is carried out using batch mode whereby cleansed data in CSV formats are imported into HDFS using Hadoop file system commands. Before ingestion, preprocessing like null values remove, fixing formatting errors and standardizing column structures are implemented. This guarantees compatibility with MapReduce processing and limits chances of having runtime errors.

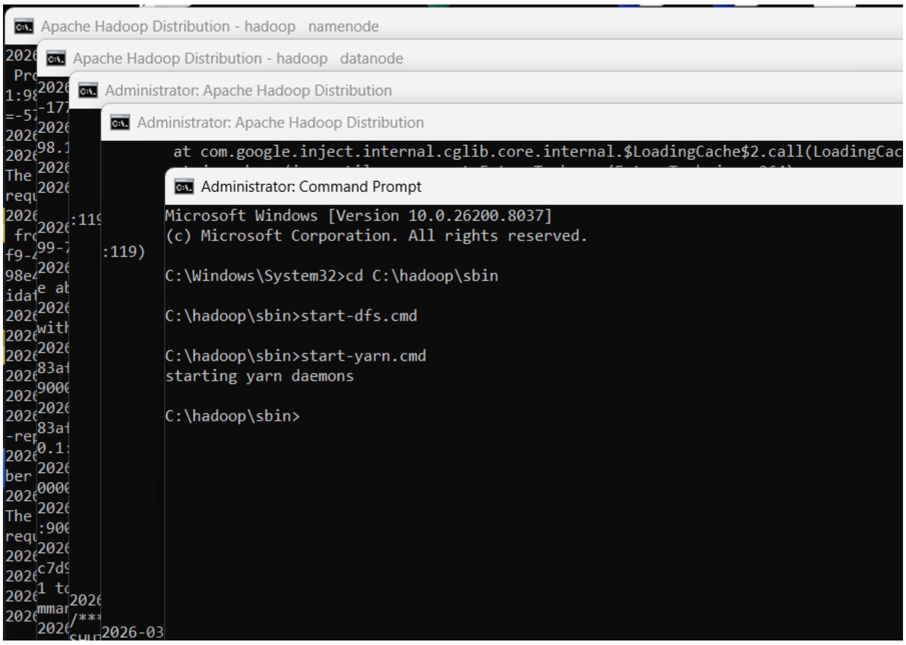


Figure 3: Hadoop Environment Initialization Using Command Prompt

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>cd C:\hadoop\sbin

C:\hadoop\sbin>start-dfs.cmd

C:\hadoop\sbin>start-yarn.cmd
starting yarn daemons

C:\hadoop\sbin>hadoop fs -ls /
Found 1 items
drwxr-xr-x   - User supergroup          0 2026-03-06 23:44 /user

C:\hadoop\sbin>
```

Figure 4: Starting Hadoop Services and Verifying HDFS Initialization

6.2 Data Storage and Organization (Phase 2)

This stage entails organizing and optimization of data in HDFS and MongoDB. HDFS is a storage platform of raw transactional data accessed and processed in parallel by multiple nodes using distributed trails of 128 MB block size. Data are structured in reasonable folders to enhance manageability and access control.

MongoDB will be designed as the second layer of storage of processed data. Tables like sales-data and metadata are developed to store structured outputs and information about the system. Indexing is used on key fields like product category in order to enhance the speed of queries. The dual-storage solution will guarantee that big datasets will be stored in HDFS to be processed, whereas fast data retrieval to analytics will be supported by MongoDB.

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>cd C:\hadoop\sbin

C:\hadoop\sbin>start-dfs.cmd

C:\hadoop\sbin>start-yarn.cmd
starting yarn daemons

C:\hadoop\sbin>hadoop fs -ls /
Found 1 items
drwxr-xr-x   - User supergroup          0 2026-03-06 23:44 /user

C:\hadoop\sbin>hadoop fs -mkdir -p /user/agathiiyan/assignment2/data

C:\hadoop\sbin>hadoop fs -put "C:\hadoop_lab\Assignment 2\dataset\online_retail_cleaned_1500.csv" /user/agathiiyan/assig
nment2/data/

C:\hadoop\sbin>hadoop fs -ls /user/agathiiyan/assignment2/data
Found 1 items
-rw-r--r--   1 User supergroup      83846 2026-03-24 15:04 /user/agathiiyan/assignment2/data/online_retail_cleaned_1500.
csv

C:\hadoop\sbin>
```

Figure 5: Uploading Dataset into HDFS

```
Administrator: Command Prompt
                Reduce output records=0
                Spilled Records=0
                Shuffled Maps =1
                Failed Shuffles=0
                Merged Map outputs=1
                GC time elapsed (ms)=4
                Total committed heap usage (bytes)=339738624
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=83846
        File Output Format Counters
                Bytes Written=0

C:\hadoop_lab\Assignment 2\MapReducePrograms>hadoop fs -ls /user/agathiiyan/assignment2/output_retail
Found 2 items
-rw-r--r--   1 User supergroup          0 2026-03-24 15:48 /user/agathiiyan/assignment2/output_retail/_SUCCESS
-rw-r--r--   1 User supergroup          0 2026-03-24 15:48 /user/agathiiyan/assignment2/output_retail/part-r-00000

C:\hadoop_lab\Assignment 2\MapReducePrograms>
```

Figure 6: Verification of Data Storage Structure in HDFS

6.3 MapReduce Job Development and Testing (Phase 3)

This process incorporates the development of MapReduce program to carry out sales aggregation. The implementation of the job is done in Java where Mapper class processes key-value pairs where one is product category and the other price and Reducer class sums values relating to each key.

The job is bundled together and packed in a JAR file and executed by Hadoop commands. In the execution Hadoop divides input data automatically into blocks and distributes tasks among a number of nodes. The shuffle, and sort phase process intermediate data, the data are correctly grouped before being reduced.

The testing is conducted based on sample data to measure the accuracy of the output. In performance tuning parameters that are manipulated include the number of mapper and reducer tasks. To determine the bottlenecks and efficient implementation, logs and counters are monitored.

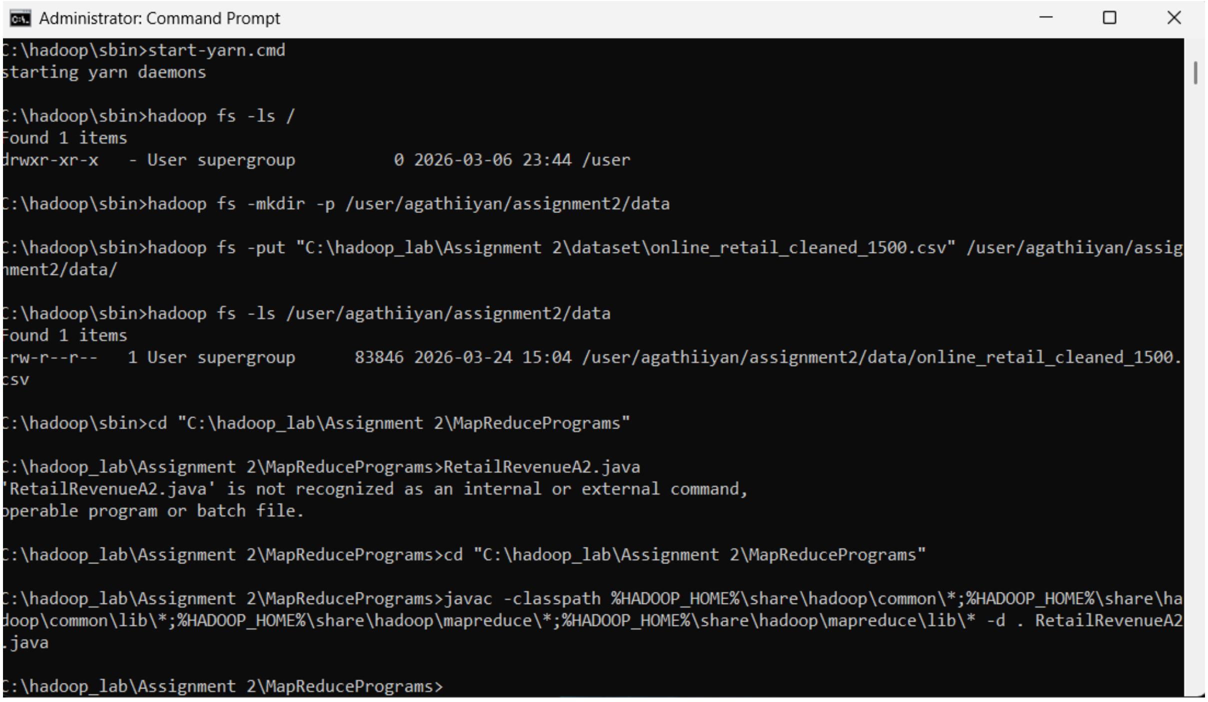


Figure 7: Compilation of MapReduce Java Program using JDK

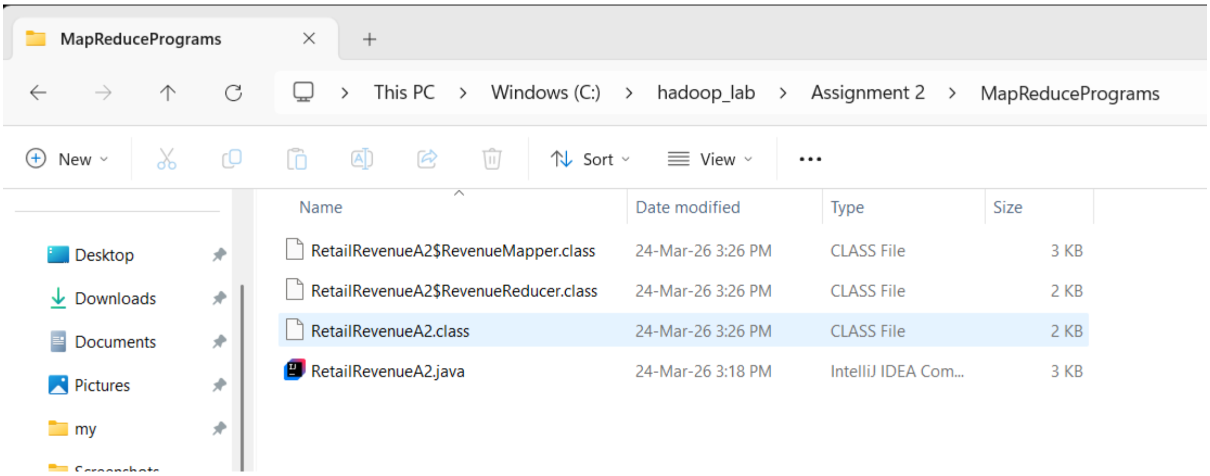


Figure 8: Generated Class Files after Successful Compilation

6.4 Analytics and Visualization Implementation (Phase 4)

The last stage is called the final step, which aims at activating the processed data to be used in business. The MapReduce output that is placed in HDFS (part-r-00000 files) is moved into MongoDB via data integration tools or scripts. This makes it possible to have an organized storage of aggregated results, which can be easily queried.

MongoDB is the source of analytics tools, which includes Power BI, in which dashboards can be created by visualizing the total sales per line of products sold. The indexing and aggregation pipelines of MongoDB are optimized to conduct query operations and ensure that they respond quickly.

Even though the implementation of full visualization might not be a solution of the current project, the design of the architecture facilitates seamless integration with outside analytics systems in order to provide real-time business data within a production setup.

```
Administrator: Command Prompt
C:\hadoop_lab\Assignment 2\MapReducePrograms>hadoop fs -cat /user/agathiiyan/assignment2/output_retail/part-r-00000
"CHARLIE+LOLA""EXTREMELY BUSY"" SIGN"   2.55
"RECORD FRAME 7"" SINGLE SIZE " 4.65
10 COLOUR SPACEBOY PEN  1.7
12 DAISY PEGS IN WOOD BOX       8.25
12 MESSAGE CARDS WITH ENVELOPES 1.65
12 PENCIL SMALL TUBE WOODLAND   0.65
12 PENCILS SMALL TUBE SKULL     0.65
12 PENCILS TALL TUBE SKULLS     0.85
20 DOLLY PEGS RETROSPOT 1.25
200 RED + WHITE BENDY STRAWS    1.25
3 HOOK HANGER MAGIC GARDEN      1.95
3 HOOK PHOTO SHELF ANTIQUE WHITE        8.5
3 PIECE SPACEBOY COOKIE CUTTER SET      4.2
3 STRIPEY MICE FELTCRAFT        5.85
3 TIER CAKE TIN GREEN AND CREAM 44.849999999999994
3 TIER CAKE TIN RED AND CREAM   14.95
36 FOIL HEART CAKE CASES        6.300000000000001
36 FOIL STAR CAKE CASES 2.1
36 PENCILS TUBE RED RETROSPOT   1.25
36 PENCILS TUBE WOODLAND        1.25
3D CHRISTMAS STAMPS STICKERS    2.5
3D DOG PICTURE PLAYING CARDS    2.95
4 IVORY DINNER CANDLES SILVER FLOCK     2.55
4 PURPLE FLOCK DINNER CANDLES   2.55
4 TRADITIONAL SPINNING TOPS     2.5
5 HOOK HANGER MAGIC TOADSTOOL   4.949999999999999
5 HOOK HANGER RED MAGIC TOADSTOOL       6.6
5 STRAND GLASS NECKLACE CRYSTAL 19.049999999999997
6 RIBBONS ELEGANT CHRISTMAS     1.65
6 RIBBONS RUSTIC CHARM  6.6
6 RIBBONS SHIMMERING PINKS      1.65
60 CAKE CASES DOLLY GIRL DESIGN 0.55
60 CAKE CASES VINTAGE CHRISTMAS 1.1
60 TEATIME FAIRY CAKE CASES     2.2
72 SWEETHEART FAIRY CAKE CASES  1.6500000000000001
ADVENT CALENDAR GINGHAM SACK    5.95
AGED GLASS SILVER T-LIGHT HOLDER        2.5
AIRLINE BAG VINTAGE JET SET WHITE       4.25
AIRLINE BAG VINTAGE TOKYO 78    4.25
AIRLINE BAG VINTAGE WORLD CHAMPION      4.25
AIRLINE LOUNGE-METAL SIGN       2.1
ALARM CLOCK BAKELIKE GREEN      18.75
ALARM CLOCK BAKELIKE IVORY      11.25
ALARM CLOCK BAKELIKE ORANGE     3.75
ALARM CLOCK BAKELIKE PINK       3.75
ALARM CLOCK BAKELIKE RED        15.0
ANGEL DECORATION STARS ON DRESS 0.42
ANTIQUE GLASS DRESSING TABLE POT        5.9
```

Figure 9: MapReduce Output Stored in HDFS (part-r-00000 File)

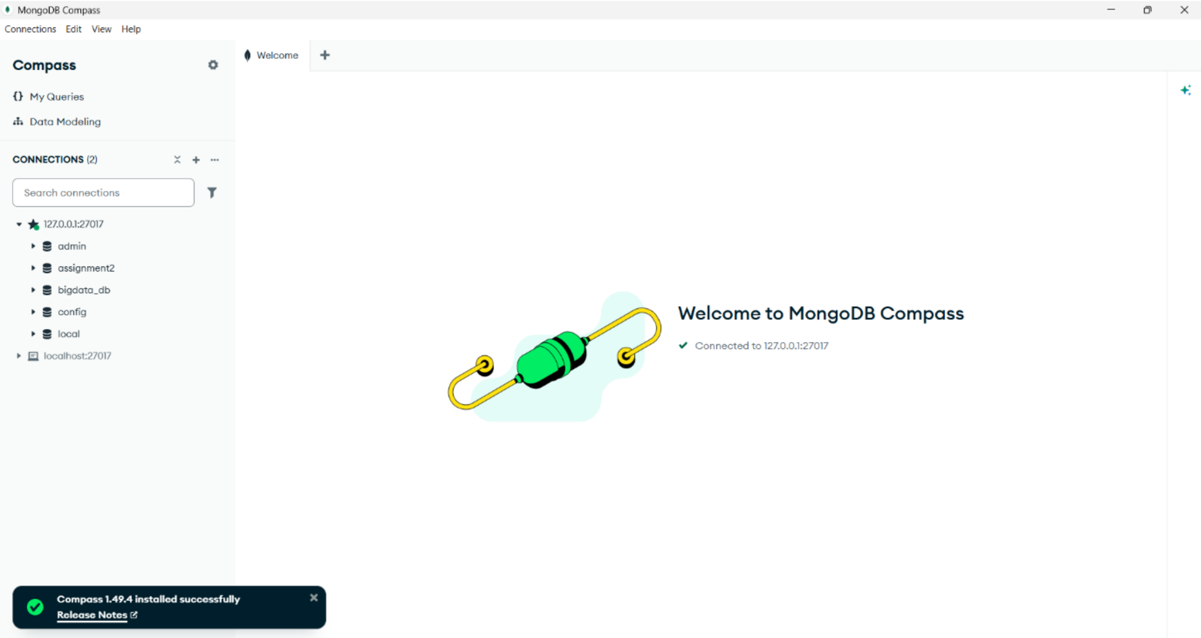


Figure 10: MongoDB Compass Connection Established

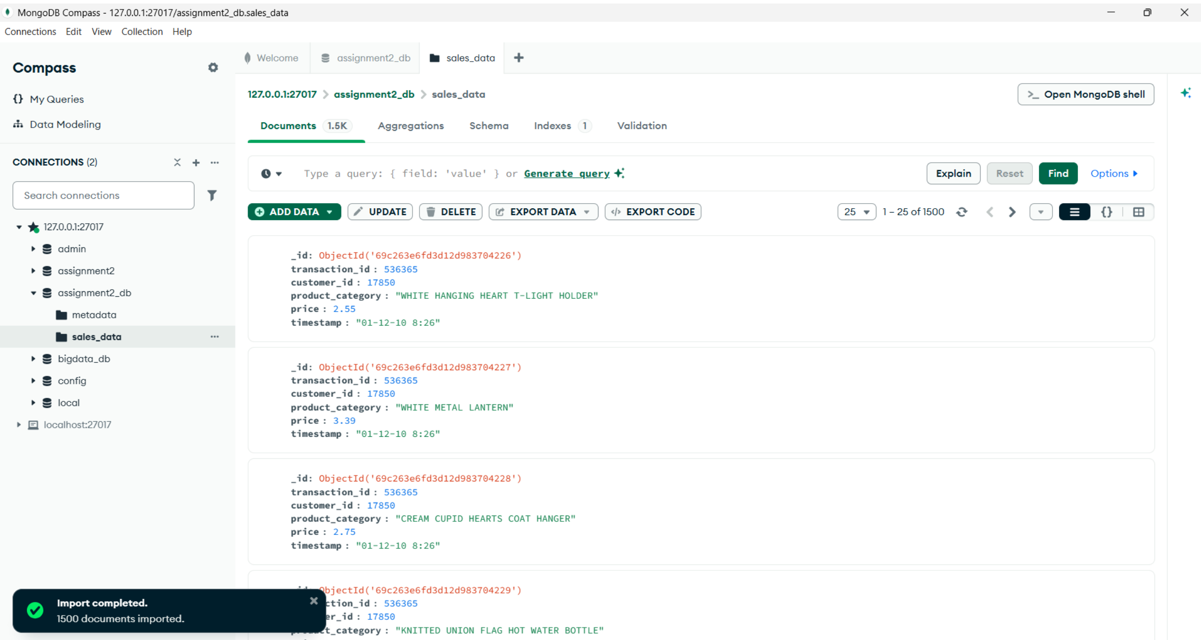


Figure 11: Processed Data Imported into MongoDB Collection

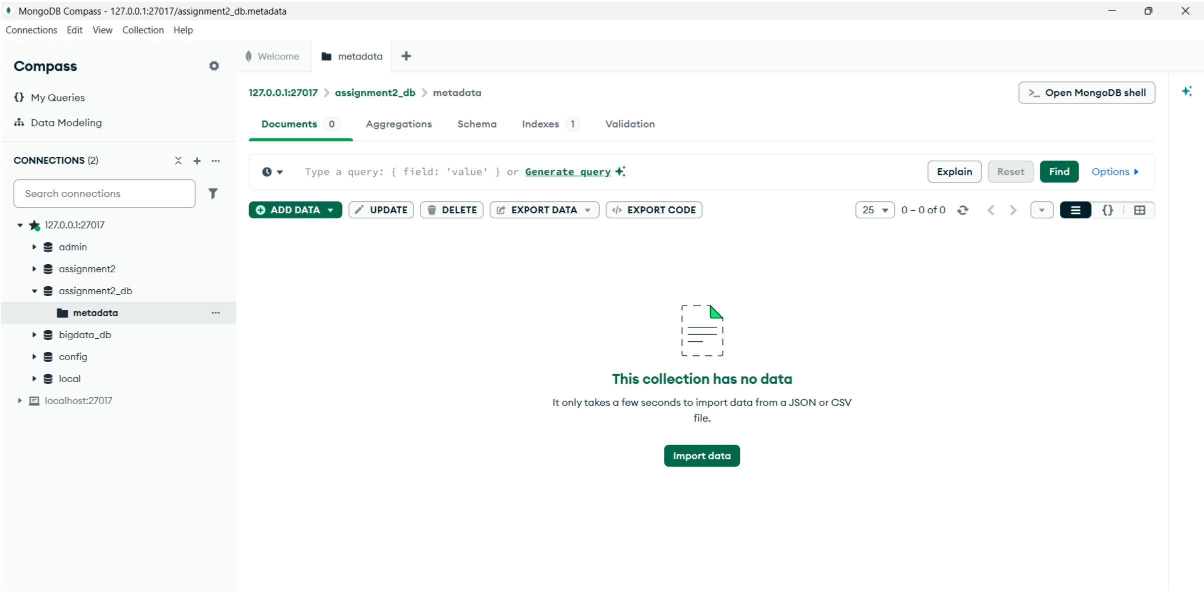


Figure 12: MongoDB Database and Collections Structure

Phase	Description	Duration	Key Deliverables
Phase 1	Infrastructure setup and data ingestion	2 weeks	Hadoop cluster configured, data uploaded to HDFS
Phase 2	Data storage and organization	1 week	Data structured in HDFS and MongoDB
Phase 3	MapReduce development and testing	3 weeks	MapReduce job completed and validated
Phase 4	Analytics and visualization	2 weeks	Data ready for reporting and dashboards

Table 1: Implementation Phases and Project Timeline for NovaRetail Big Data System

6.5 System Implementation Details

The system implementation was done on one-node Hadoop cluster deployed on a windows-based platform. Though, it is implemented on one computer, the configuration mimics a distributed computing setting in which all Hadoop services, such as the NameNode, DataNode,ResourceManager, and NodeManager, are co-located to execute storage, and processing operations.

The command-line scripts were used to start the Hadoop services which means that the HDFS and YARN services were started properly before starting any processing tasks. The process monitoring commands were used to do system verification to ensure that all the necessary services were up and running as desired.

The data was put into HDFS with the Hadoop file system instructions and successful storing was checked through directory listing instructions. This made the data available to MapReduce processing. The data were kept in organized directories to ensure that there is orderliness and facilitate easy retrieval of data.

The MapReduce software has been written in Java and it is compiled with Java Development Kit (JDK). The resulting class files were put together in an executable JAR file, and this was executed in the Hadoop environment. The implementation process entailed the definition of input and output paths in the HDFS enabling the use of Hadoop to process the data in a distributed fashion.

In the process of execution, Hadoop was automatic in the division of data, distribution of tasks and parallel tasks. The HDFS output directory provided the final output and was later checked using Hadoop commands to be sure that results were correct and complete.

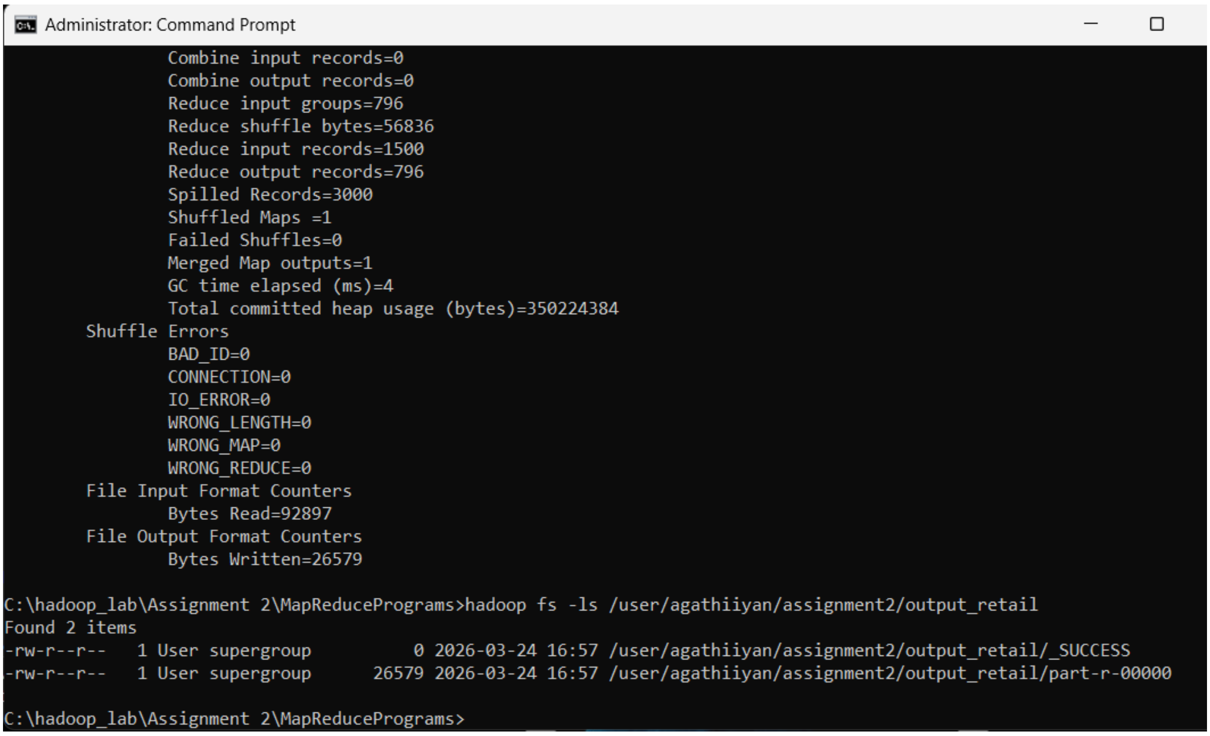


Figure 13: Final Output Directory Generated After MapReduce Execution

```
Administrator: Command Prompt
ANTIQUE GLASS HEART DECORATION  1.95
ANTIQUE GLASS PEDESTAL BOWL     3.75
ANTIQUE SILVER TEA GLASS ENGRAVED       1.06
ANTIQUE TALL SWIRLGLASS TRINKET POT     3.75
APPLE BATH SPONGE       1.25
ASS COL SMALL SAND GECKO P'WEIGHT       0.42
ASSORTED BOTTLE TOP  MAGNETS    0.42
ASSORTED COLOUR BIRD ORNAMENT   13.519999999999998
ASSORTED COLOUR LIZARD SUCTION HOOK     0.42
ASSORTED COLOUR MINI CASES      23.85
BABUSHKA LIGHTS STRING OF 10    27.0
BAG 500g SWIRLY MARBLES 1.65
BALLOON ART MAKE YOUR OWN FLOWERS       1.95
BALLOONS  WRITING SET   3.3
BATH BUILDING BLOCK WORD        5.95
BATHROOM METAL SIGN     0.55
BIRD DECORATION RED RETROSPOT   0.85
BIRD HOUSE HOT WATER BOTTLE     5.1
BIRTHDAY CARD- RETRO SPOT       0.42
BIRTHDAY PARTY CORDON BARRIER TAPE      1.25
BISCUIT TIN VINTAGE RED 6.75
BLACK CANDELABRA T-LIGHT HOLDER 4.2
BLACK DIAMANTE EXPANDABLE RING  4.25
BLACK DINER WALL CLOCK  8.5
BLACK HEART CARD HOLDER 7.34
BLACK KITCHEN SCALES    8.5
BLACK LOVE BIRD CANDLE  2.5
BLACK ORANGE SQUEEZER   0.42
BLACK PIRATE TREASURE CHEST     1.65
BLACK RECORD COVER FRAME        3.39
BLACK SWEETHEART BRACELET       4.25
BLACK TEA TOWEL CLASSIC DESIGN  2.5
BLACK/BLUE POLKADOT UMBRELLA    17.85
BLUE 3 PIECE POLKADOT CUTLERY SET       7.5
BLUE BIRDHOUSE DECORATION       0.85
BLUE CHARLIE+LOLA PERSONAL DOORSIGN     2.95
BLUE COAT RACK PARIS FASHION    9.9
BLUE DINER WALL CLOCK   17.0
BLUE DRAGONFLY HELICOPTER       7.95
BLUE DRAWER KNOB ACRYLIC EDWARDIAN      1.25
BLUE GIANT GARDEN THERMOMETER   5.95
BLUE HARMONICA IN BOX   1.25
BLUE NEW BAROQUE CANDLESTICK CANDLE     8.850000000000001
BLUE OWL SOFT TOY       2.95
BLUE PAISLEY NOTEBOOK   1.65
BLUE PAISLEY TISSUE BOX 2.5
BLUE PAPER PARASOL      2.95
BLUE PARTY BAGS 2.1
BLUE POLKADOT PLATE     1.69
```

Figure 14: Verified Aggregated Sales Results from HDFS

7.0 Challenges and Solutions

The deployment of a big data solution to NovaRetail Analytics Sdn. Bhd. is associated with a number of technical and operational issues. All these challenges should be considered so that scalability, performance, and reliability of the system are realized. This part describes the major issues that arise and the solutions that have been undertaken in the proposed architecture.

7.1 Scalability Challenges

With the continued expansion of NovaRetail, the amount of transactional data to be processed is likely to swell, particularly at peak times of the business like major promotional campaigns. Unless nurtured through proper mechanisms, this increased rate of data might consume the resources of the system and influence system levels of processing.

In order to overcome this problem, the structure utilizes the distributed computing of Hadoop where extra nodes can be added to the cluster at the demand. This process of horizontal scaling will guarantee the storage and processing capacity can be enlarged according to the growth of data. In addition, HDFS will automatically distribute the data among the nodes to provide the capability of running processes in parallel and keep the system running.

7.2 Data Quality and Consistency Issue

The data that has been gathered using various sources usually has discrepancies, missing values or format errors. Such problems may give wrong analytical conclusions and minimize the accuracy of the insights formed by the system. In order to avert this, data preprocessing and validation measures are introduced prior to being ingested in HDFS. Intervention methods like data cleaning, normalization and format standardization methods are done to provide uniformity. Also, we have validation rules that are used to identify and process anomalies so that only quality data is processed through the system.

7.3 Security and Privacy Concern

The data on a large number of customers poses major security and privacy threats. Illegal invasion or security failures might steal confidential customer data and contravene the regulatory standards like the Personal Data Protection Act (PDPA) in Malaysia. To overcome these issues access control mechanisms are introduced in the Hadoop ecosystem such that access to data is limited to authorized people. Protection of sensitive information can also be done through encryption of data during storage and transmission. Role Based Access Control (RBAC) is employed in Mongo database in order to control the permissions of the users and thereby handle data securely.

7.4 Performance Optimization Challenges

The time processing large datasets with MapReduce may take is longer without proper optimization. Inefficient job configuration, imbalanced data distribution, as well as overworked disk I/O can affect the performance. A number of optimization strategies are used in order to enhance efficiency. They are the tuning of the number of mapper and reducer tasks, the balancing of data partitioning, and the use of the data locality to have reduced network overhead. Moreover, the compression can be applied to intermediate data too, which will be accomplished during the shuffle phase and will minimize the time spent on transferring data and enhance the performance.

7.5 System Configuration and Implementation Issues

Various pitfalls were experienced during the implementation process in regards to the system setup and implementation in the Hadoop environment. Problems with the environment variables being set incorrectly, port conflicts, and service startup failures for example the NameNode or ResourceManager not starting at all interfered with the performance of the successful execution of MapReduce jobs.

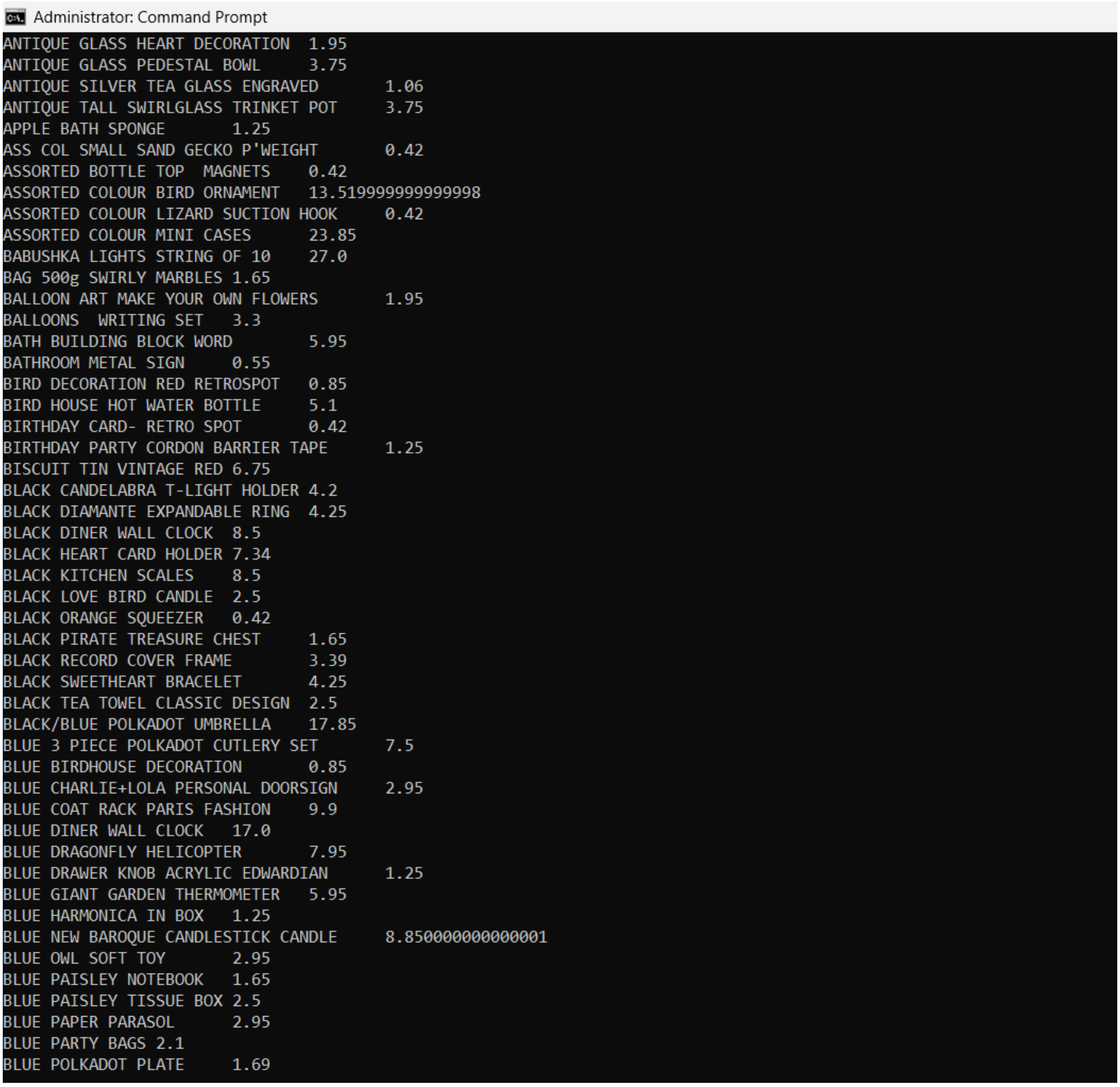
Also, connection failure like an error of connection refused was reported when connecting to HDFS such as localhost:9000. This implies that Hadoop services had not been configured or started. Such problems may cause disruption in the whole process of work, particularly when it comes to uploading data or carrying out processing activities.

In order to overcome these issues, the environment variables like JAVAHOME, HADOOPHOME and system PATH were configured properly. The correct scripts like start-dfs.cmd, start-yarn.cmd, stop-dfs.cmd, stop-yarn.cmd were used to start and stop the Hadoop services in order to ensure stability of the system. The techniques of troubleshooting which included checking running processes with jps, checking configuration files and restarting services were implemented to make sure the MapReduce jobs ran successfully.

8.0 Results and Analysis

The implementation of the MapReduce program was performed efficiently to produce output results which reflects the sum in sales per category of product based on the transaction data. The findings were saved in the Hadoop Distributed File system (HDFS) and were subsequently incorporated into MongoDB where they would be stored and analyzed.

The result proves that the MapReduce system could effectively compute the data with the help of distributed computation. The different product categories were merged properly and came up with significant findings on sales performance. This proves that Hadoop is effective in processing transactional data in large volumes.



```
Administrator: Command Prompt
ANTIQUE GLASS HEART DECORATION  1.95
ANTIQUE GLASS PEDESTAL BOWL     3.75
ANTIQUE SILVER TEA GLASS ENGRAVED       1.06
ANTIQUE TALL SWIRLGLASS TRINKET POT     3.75
APPLE BATH SPONGE       1.25
ASS COL SMALL SAND GECKO P'WEIGHT       0.42
ASSORTED BOTTLE TOP  MAGNETS    0.42
ASSORTED COLOUR BIRD ORNAMENT   13.519999999999998
ASSORTED COLOUR LIZARD SUCTION HOOK     0.42
ASSORTED COLOUR MINI CASES      23.85
BABUSHKA LIGHTS STRING OF 10    27.0
BAG 500g SWIRLY MARBLES 1.65
BALLOON ART MAKE YOUR OWN FLOWERS       1.95
BALLOONS  WRITING SET   3.3
BATH BUILDING BLOCK WORD        5.95
BATHROOM METAL SIGN     0.55
BIRD DECORATION RED RETROSPOT   0.85
BIRD HOUSE HOT WATER BOTTLE     5.1
BIRTHDAY CARD- RETRO SPOT       0.42
BIRTHDAY PARTY CORDON BARRIER TAPE      1.25
BISCUIT TIN VINTAGE RED 6.75
BLACK CANDELABRA T-LIGHT HOLDER 4.2
BLACK DIAMANTE EXPANDABLE RING  4.25
BLACK DINER WALL CLOCK  8.5
BLACK HEART CARD HOLDER 7.34
BLACK KITCHEN SCALES    8.5
BLACK LOVE BIRD CANDLE  2.5
BLACK ORANGE SQUEEZER   0.42
BLACK PIRATE TREASURE CHEST     1.65
BLACK RECORD COVER FRAME        3.39
BLACK SWEETHEART BRACELET       4.25
BLACK TEA TOWEL CLASSIC DESIGN  2.5
BLACK/BLUE POLKADOT UMBRELLA    17.85
BLUE 3 PIECE POLKADOT CUTLERY SET       7.5
BLUE BIRDHOUSE DECORATION       0.85
BLUE CHARLIE+LOLA PERSONAL DOORSIGN     2.95
BLUE COAT RACK PARIS FASHION    9.9
BLUE DINER WALL CLOCK   17.0
BLUE DRAGONFLY HELICOPTER       7.95
BLUE DRAWER KNOB ACRYLIC EDWARDIAN      1.25
BLUE GIANT GARDEN THERMOMETER   5.95
BLUE HARMONICA IN BOX   1.25
BLUE NEW BAROQUE CANDLESTICK CANDLE     8.850000000000001
BLUE OWL SOFT TOY       2.95
BLUE PAISLEY NOTEBOOK   1.65
BLUE PAISLEY TISSUE BOX 2.5
BLUE PAPER PARASOL      2.95
BLUE PARTY BAGS 2.1
BLUE POLKADOT PLATE     1.69
```

Figure 15: Sample Output Showing Total Sales Per Product

8.1 MapReduce Output Verification

Upon completion of the MapReduce job, the standard output files such as the SUCCESS and part-r-00000 were found in the output directory in HDFS. The SUCCESS file shows that the job has been accomplished successfully and part-r-00000 file contains the actual results of the process.

The output was verified using Hadoop commands such as:

- hadoop fs -ls /output_retail
- hadoop fs -cat /output_retail/part-r-00000

The findings showed category of products and their respective total sales values which proved that the reducer was working as expected to combine the data.

8.2 Sample Output Interpretation

A sample of the output is shown below:

Product Category	**Total Sales (RM)**
WHITE HANGING HEART T-LIGHT HOLDER	2.55
WHITE METAL LANTERN	3.39
CREAM CUPID HEARTS COAT HANGER	2.75

Table 2: Sample Output of Total Sales per Product Category

Each row represents a product category and its total revenue calculated from multiple transactions. The results are derived using the formula:

→Total Sales = Quantity × Unit Price (aggregated across transactions)

This confirms that the MapReduce logic is functioning correctly and producing accurate aggregation results.

8.3 Business Insights from Analysis

To make a business decision, the information on the analysis of total sales per product category is valuable. The products that have better total sale counts show the demand of the products by the customers and can be considered when managing inventories and promotional strategies. Indicatively, common-selling products like home decoration materials, and gifted products indicate steady contribution to revenue as most of them sell regularly indicating that the products are major sales drivers.

Conversely, less selling products might need promotion techniques or price change in order to enhance performance. These observations reveal that the use of big data analytics can facilitate the use of data in the decision-making process in the e-commerce context to help businesses maximize their operations to enhance profitability.

8.4 MongoDB Data Validation

The results of the processing were successfully stored in MongoDB collections and further analysis and querying were possible. Sales data collection will have information on the transactions of the business in a structured form and metadata information will be stored in a separate collection. Authentication was carried out through MongoDB Compass where documents were viewed in order to ensure that data was properly imported.

The fact that there were several records and that the fields had been organized shows that there was successful integration between Hadoop and MongoDB. This integration makes the system better by allowing faster data access and allows real-time analytics use cases to be effectively served, which are not currently well served by HDFS itself.

9.0 System Evaluation and Future Enhancement

The deployed big data architecture of NovaRetail Analytics Sdn. Bhd. is evidence of the ability to manage transactional data effectively with the distributed storage and processing technology. The system combines Hadoop, MapReduce, and MongoDB and is effective in processing and analysis of retail data to give meaningful insights on total sales on a product category.

Technically, the system demonstrates good performance when performing batch data processing. The distributed framework of Hadoop enables parallel processing of data and executing it at a shorter period as compared to the traditional sequential processes. Although the system is deployed on a single-node environment, it serves as a good simulation to a distributed architecture, and it shows how scalability is possible when the system is deployed in real-life settings.

There were a few limitations however that were realized through the implementation. A single-node cluster can only limit the full performance of Hadoop because the ability to get parallelism in multiple machines is not entirely realized. Moreover, MapReduce processing is slow, though dependable, in comparison with the contemporary in-memory processing frameworks.

A number of improvements can be suggested to make the system even better. Migration to a multi-node Hadoop cluster as opposed to a single node would be one of the possible improvements, and this would dramatically increase the processing speed and scalability. One more redesign is the addition of Apache Spark that is capable of in-memory processing data faster and has real-time analytics potential.

Moreover, automated data pipelines can be added to the system with the help of such tools as Apache Kafka or Apache NiFi that can be used to receive real-time data. This would make it possible to have continuous data processing as opposed to using only the batch processing method.

Regarding business, the combination of advanced analytics and visualization tools like Power BI can also increase the levels of decision-making. Interactive dashboards may offer businesses a real-time view of sales trends, customer behavior, and product performance so that companies can react swiftly to modifications in the market.

On the whole, the system is capable of forming a good basis of big data analytics within an e-commerce setting. It can become an all-production ready analytics platform that is able to deal with the large-scale business operation with further improvements and the incorporation of scalability.

Aspect	**Current Implementation**	**Future Enhancement**
Cluster Setup	Single-node Hadoop	Multi-node distributed cluster
Processing	MapReduce (batch)	Apache Spark (real-time)
Data Ingestion	Manual upload (HDFS)	Automated pipeline (Kafka/NiFi)
Storage	HDFS + MongoDB	Cloud storage integration
Analytics	Basic output	Interactive dashboards (Power BI)

Table 3: Comparative Analysis of Existing Architecture and Proposed Future Enhancements

10.0 Conclusion

The present project proves successful in showing how big data technologies can be applied to the analysis of transactional data in the framework of an e-commerce setting. The system could have effectively processed and stored massive amounts of data and provided meaningful information on total sales per product category by developing Hadoop, HDFS, MapReduce, and MongoDB.

The implementation brings out the success of distributed computing in data intensive tasks. By becoming a fundamental concept using MapReduce, the dataset was handled in an organized and scalable way permitting precise aggregation of sales information. HDFS was used to guarantee quality and fault-tolerant storage, and MongoDB was used to guarantee a flexible and efficient storage platform of processed results and allow retrieving the data faster.

In practical terms, the project suggests how big data analytics can be used to make business decisions. The sales performance analysis helps organizations to determine the high-selling products, efficient inventory techniques, and enhance the overall performance of the operations. This corresponds to the practical value of decision-making through data in contemporary e-commerce companies.

Though a single-node Hadoop cluster was used to implement the system, it manages to simulate a distributed environment and offers a good base of scalability. The system can be expanded into a full-fledged enterprise-level analytics platform with additional improvements like multi-node deployment, real-time processing structures and advanced visualization solutions.

To sum up, this project confirms the potential of big data technologies to turn raw transactional data into useful insights. It shows the technical capabilities and practical importance of distributed data processing which makes it a useful solution to organizations that need to use big data to gain competitive advantage.

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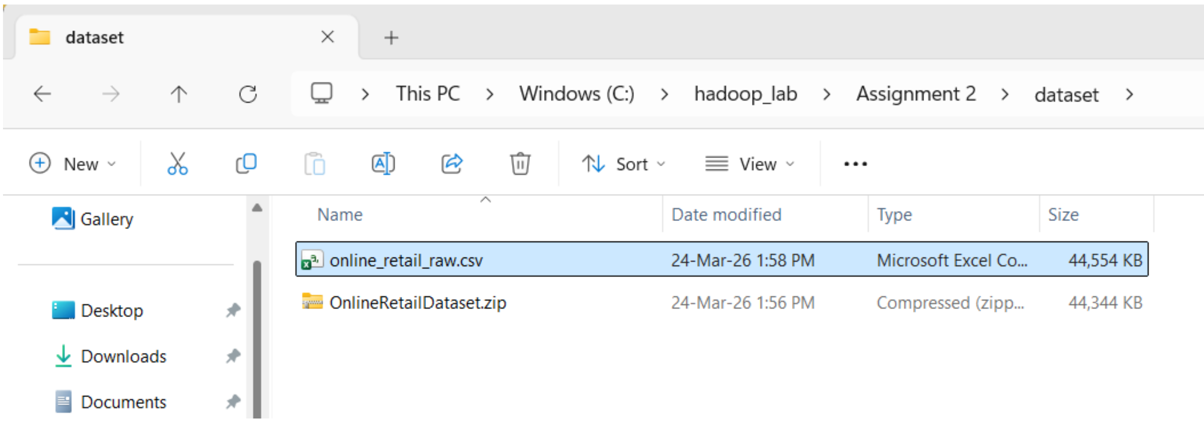
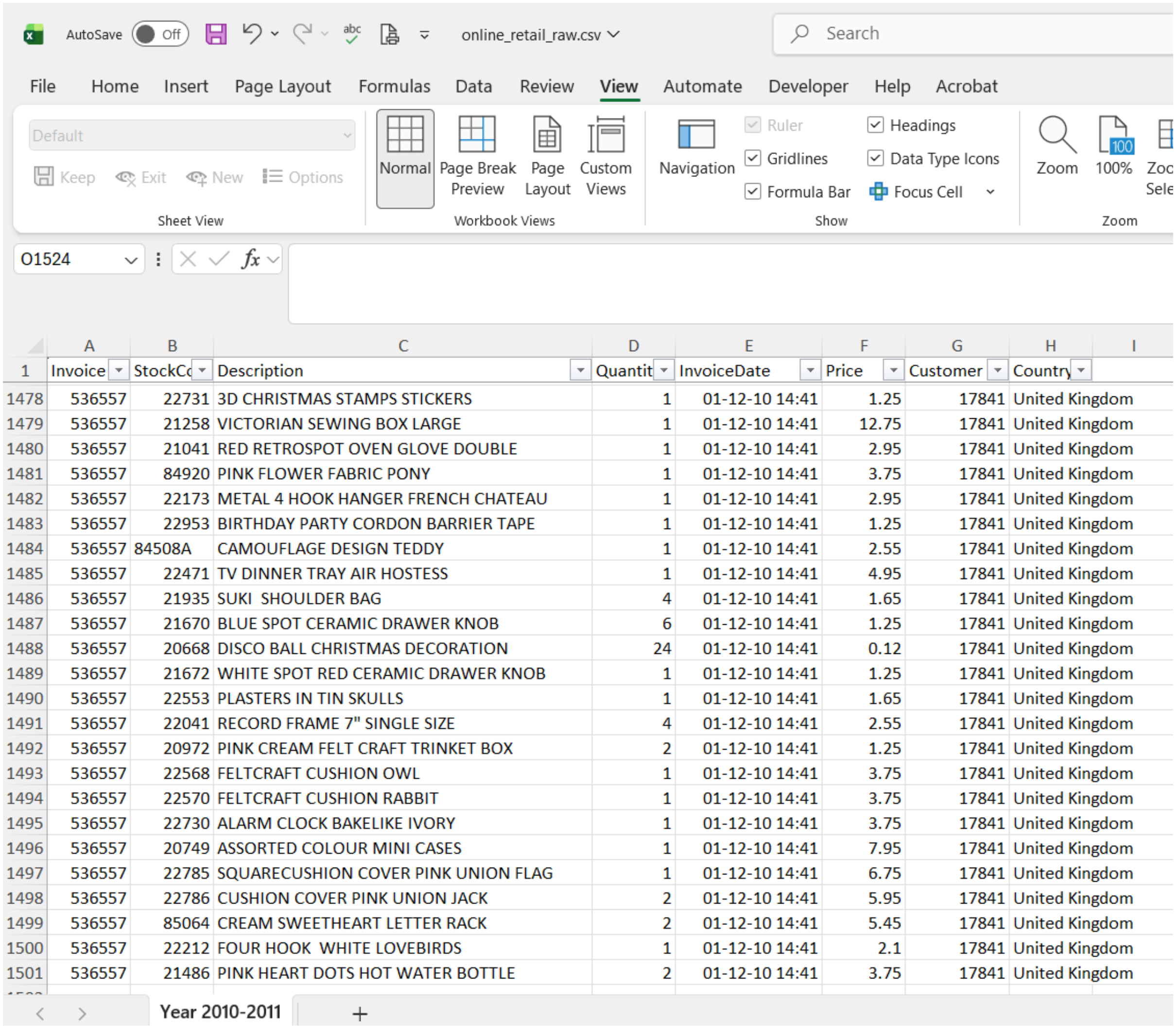
12.0 Appendix

Part 1:- Dataset Preparation

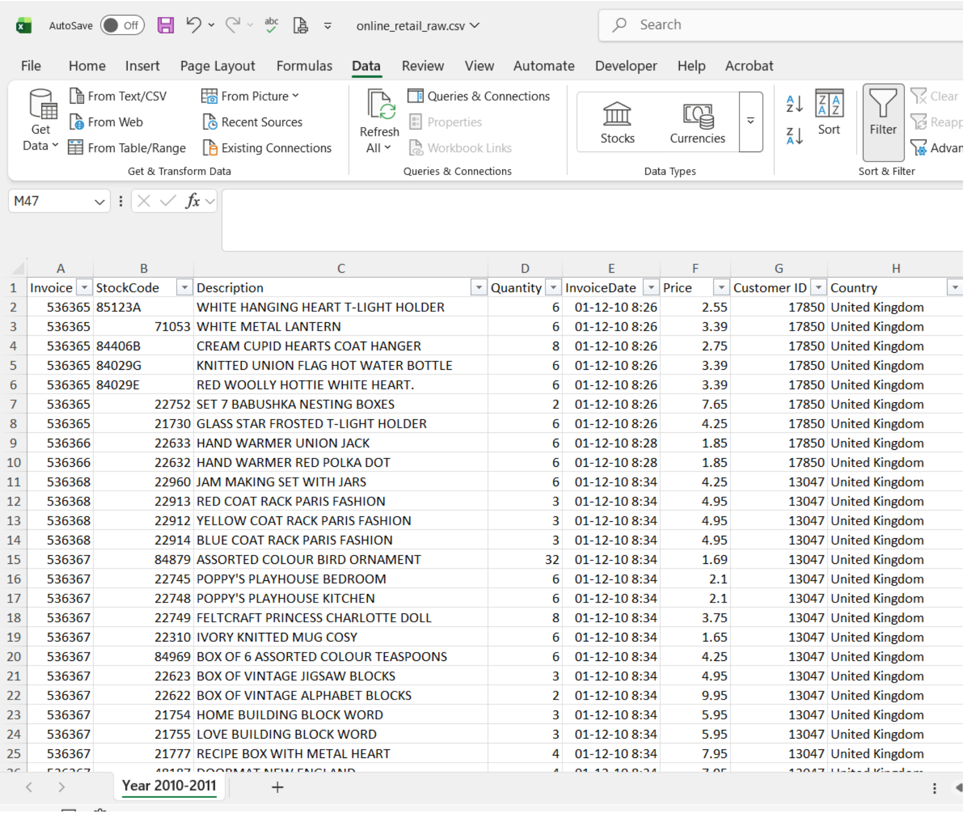


Appendix 1: Dataset Downloaded from Source



	Invoice	StockCc	Description	Quantit	InvoiceDate	Price	Customer	Country
1478	536557	22731	3D CHRISTMAS STAMPS STICKERS	1	01-12-10 14:41	1.25	17841	United Kingdom
1479	536557	21258	VICTORIAN SEWING BOX LARGE	1	01-12-10 14:41	12.75	17841	United Kingdom
1480	536557	21041	RED RETROSPOT OVEN GLOVE DOUBLE	1	01-12-10 14:41	2.95	17841	United Kingdom
1481	536557	84920	PINK FLOWER FABRIC PONY	1	01-12-10 14:41	3.75	17841	United Kingdom
1482	536557	22173	METAL 4 HOOK HANGER FRENCH CHATEAU	1	01-12-10 14:41	2.95	17841	United Kingdom
1483	536557	22953	BIRTHDAY PARTY CORDON BARRIER TAPE	1	01-12-10 14:41	1.25	17841	United Kingdom
1484	536557	84508A	CAMOUFLAGE DESIGN TEDDY	1	01-12-10 14:41	2.55	17841	United Kingdom
1485	536557	22471	TV DINNER TRAY AIR HOSTESS	1	01-12-10 14:41	4.95	17841	United Kingdom
1486	536557	21935	SUKI SHOULDER BAG	4	01-12-10 14:41	1.65	17841	United Kingdom
1487	536557	21670	BLUE SPOT CERAMIC DRAWER KNOB	6	01-12-10 14:41	1.25	17841	United Kingdom
1488	536557	20668	DISCO BALL CHRISTMAS DECORATION	24	01-12-10 14:41	0.12	17841	United Kingdom
1489	536557	21672	WHITE SPOT RED CERAMIC DRAWER KNOB	1	01-12-10 14:41	1.25	17841	United Kingdom
1490	536557	22553	PLASTERS IN TIN SKULLS	1	01-12-10 14:41	1.65	17841	United Kingdom
1491	536557	22041	RECORD FRAME 7" SINGLE SIZE	4	01-12-10 14:41	2.55	17841	United Kingdom
1492	536557	20972	PINK CREAM FELT CRAFT TRINKET BOX	2	01-12-10 14:41	1.25	17841	United Kingdom
1493	536557	22568	FELTCRAFT CUSHION OWL	1	01-12-10 14:41	3.75	17841	United Kingdom
1494	536557	22570	FELTCRAFT CUSHION RABBIT	1	01-12-10 14:41	3.75	17841	United Kingdom
1495	536557	22730	ALARM CLOCK BAKELIKE IVORY	1	01-12-10 14:41	3.75	17841	United Kingdom
1496	536557	20749	ASSORTED COLOUR MINI CASES	1	01-12-10 14:41	7.95	17841	United Kingdom
1497	536557	22785	SQUARECUSHION COVER PINK UNION FLAG	1	01-12-10 14:41	6.75	17841	United Kingdom
1498	536557	22786	CUSHION COVER PINK UNION JACK	2	01-12-10 14:41	5.95	17841	United Kingdom
1499	536557	85064	CREAM SWEETHEART LETTER RACK	2	01-12-10 14:41	5.45	17841	United Kingdom
1500	536557	22212	FOUR HOOK WHITE LOVEBIRDS	1	01-12-10 14:41	2.1	17841	United Kingdom
1501	536557	21486	PINK HEART DOTS HOT WATER BOTTLE	2	01-12-10 14:41	3.75	17841	United Kingdom

Appendix 2: Dataset Filtering to 1500 Records



Invoice	StockCode	Description	Quantity	InvoiceDate	Price	Customer ID	Country
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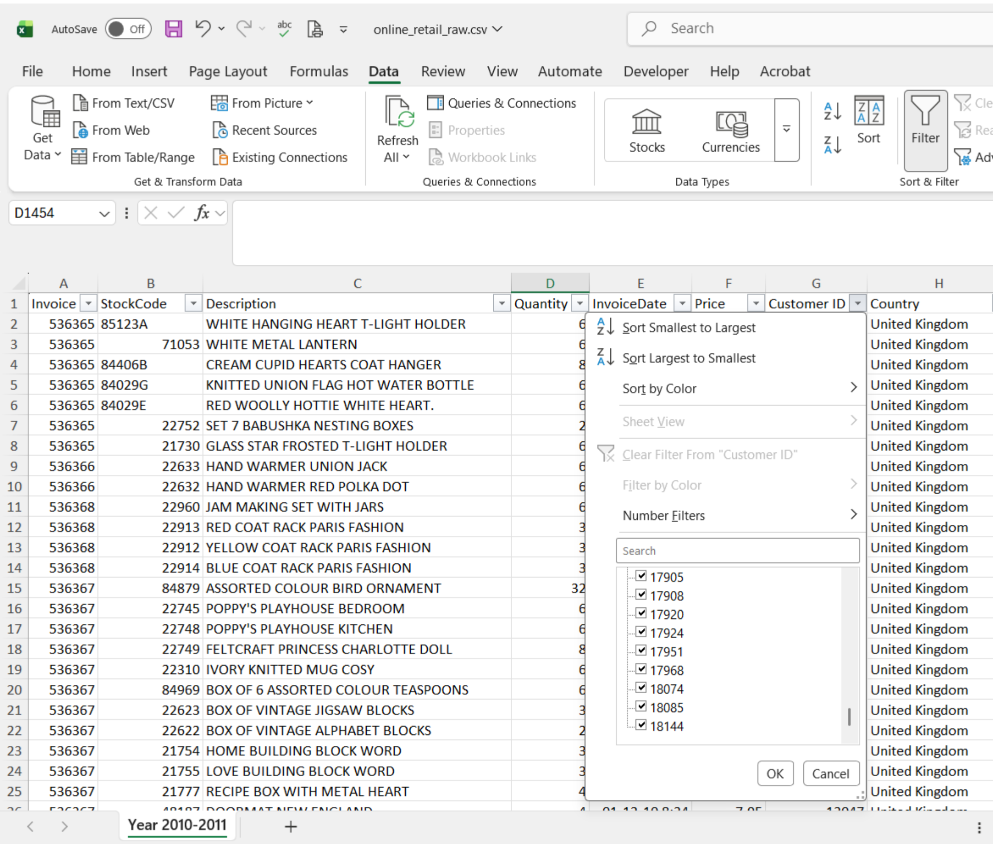
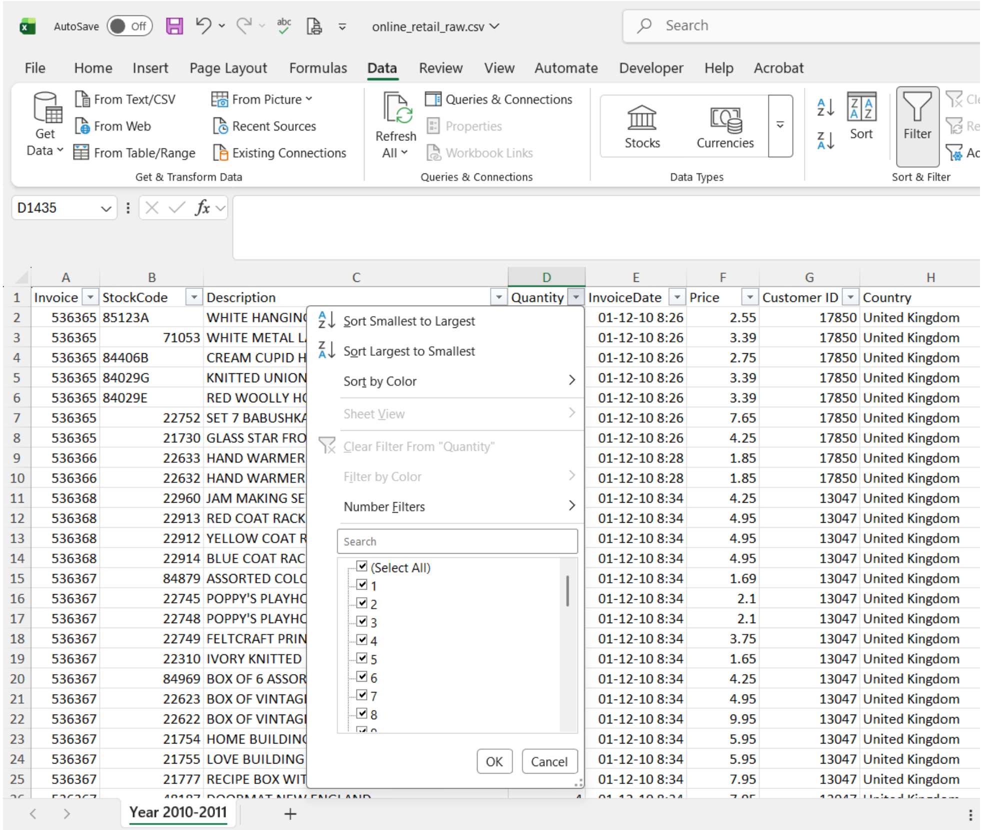
536365	71053	WHITE METAL LANTERN	6	01-12-10 8:26	3.39	17850	United Kingdom
536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	01-12-10 8:26	2.75	17850	United Kingdom
536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	01-12-10 8:26	3.39	17850	United Kingdom
536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	01-12-10 8:26	3.39	17850	United Kingdom
536365	22752	SET 7 BABUSHKA NESTING BOXES	2	01-12-10 8:26	7.65	17850	United Kingdom
536365	21730	GLASS STAR FROSTED T-LIGHT HOLDER	6	01-12-10 8:26	4.25	17850	United Kingdom
536366	22633	HAND WARMER UNION JACK	6	01-12-10 8:28	1.85	17850	United Kingdom
536366	22632	HAND WARMER RED POLKA DOT	6	01-12-10 8:28	1.85	17850	United Kingdom
536368	22960	JAM MAKING SET WITH JARS	6	01-12-10 8:34	4.25	13047	United Kingdom
536368	22913	RED COAT RACK PARIS FASHION	3	01-12-10 8:34	4.95	13047	United Kingdom
536368	22912	YELLOW COAT RACK PARIS FASHION	3	01-12-10 8:34	4.95	13047	United Kingdom
536368	22914	BLUE COAT RACK PARIS FASHION	3	01-12-10 8:34	4.95	13047	United Kingdom
536367	84879	ASSORTED COLOUR BIRD ORNAMENT	32	01-12-10 8:34	1.69	13047	United Kingdom
536367	22745	POPPY'S PLAYHOUSE BEDROOM	6	01-12-10 8:34	2.1	13047	United Kingdom
536367	22748	POPPY'S PLAYHOUSE KITCHEN	6	01-12-10 8:34	2.1	13047	United Kingdom
536367	22749	FELTCRAFT PRINCESS CHARLOTTE DOLL	8	01-12-10 8:34	3.75	13047	United Kingdom
536367	22310	IVORY KNITTED MUG COSY	6	01-12-10 8:34	1.65	13047	United Kingdom
536367	84969	BOX OF 6 ASSORTED COLOUR TEASPOONS	6	01-12-10 8:34	4.25	13047	United Kingdom
536367	22623	BOX OF VINTAGE JIGSAW BLOCKS	3	01-12-10 8:34	4.95	13047	United Kingdom
536367	22622	BOX OF VINTAGE ALPHABET BLOCKS	2	01-12-10 8:34	9.95	13047	United Kingdom
536367	21754	HOME BUILDING BLOCK WORD	3	01-12-10 8:34	5.95	13047	United Kingdom
536367	21755	LOVE BUILDING BLOCK WORD	3	01-12-10 8:34	5.95	13047	United Kingdom
536367	21777	RECIPE BOX WITH METAL HEART	4	01-12-10 8:34	7.95	13047	United Kingdom

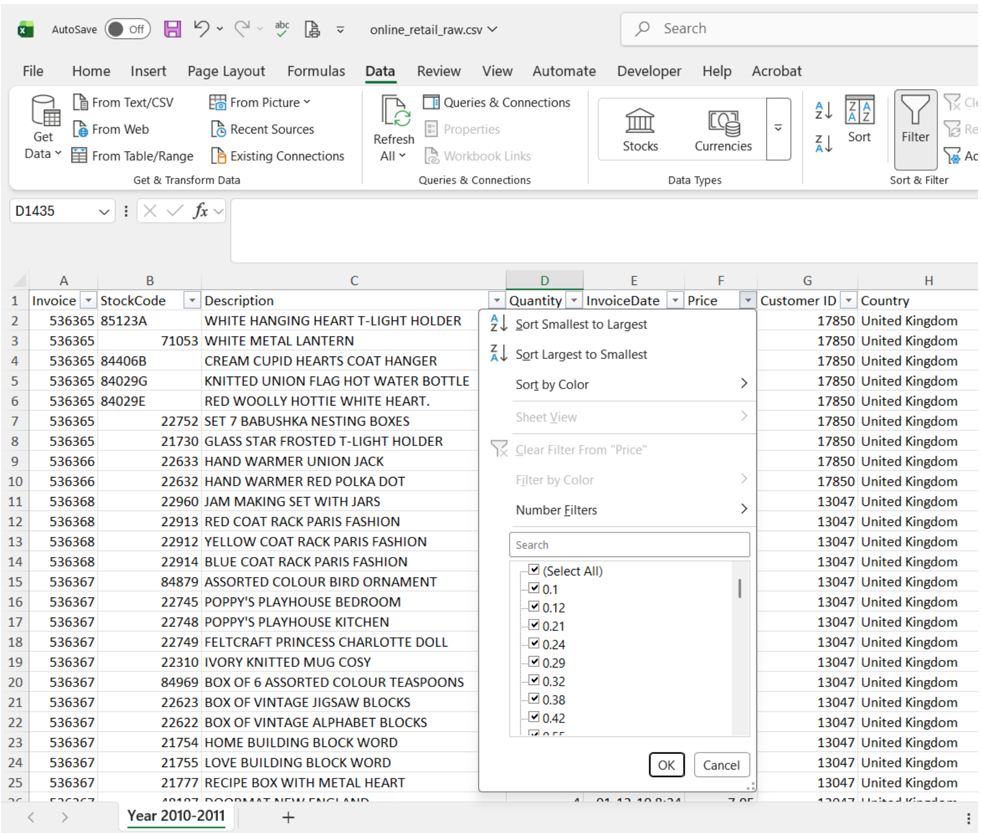
Appendix 3: Applying Data Filtering in Excel



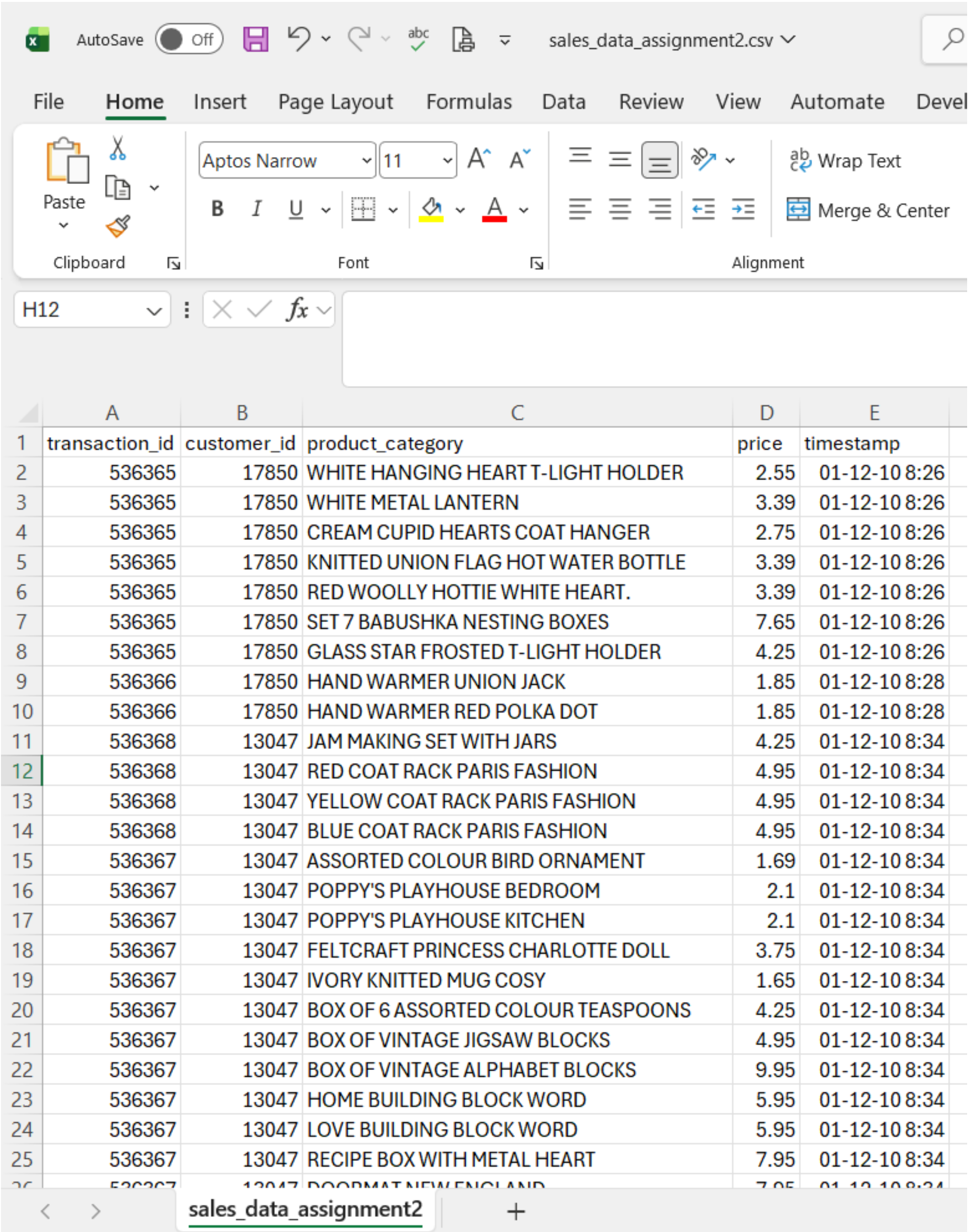
Appendix 4: Removal of Missing Customer IDs



Appendix 5: Removal of Negative Quantity Values



Appendix 6: Removal of Zero Price Records



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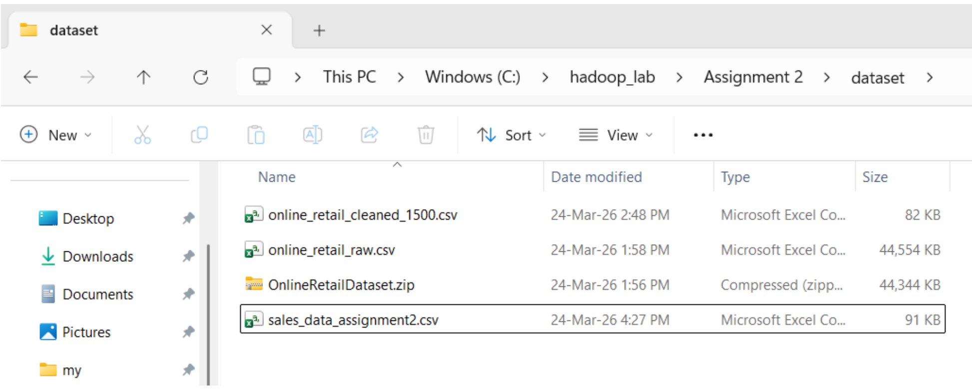
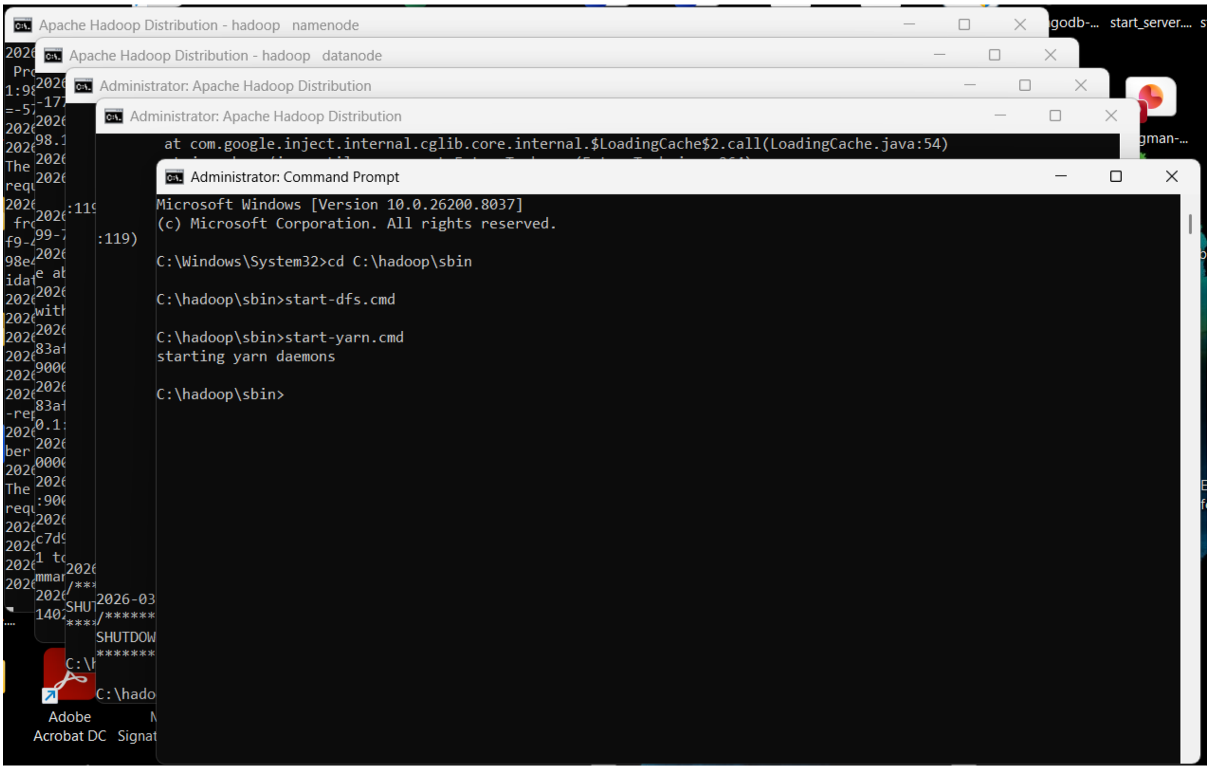
536365	17850	WHITE METAL LANTERN	3.39	01-12-10 8:26
536365	17850	CREAM CUPID HEARTS COAT HANGER	2.75	01-12-10 8:26
536365	17850	KNITTED UNION FLAG HOT WATER BOTTLE	3.39	01-12-10 8:26
536365	17850	RED WOOLLY HOTTIE WHITE HEART.	3.39	01-12-10 8:26
536365	17850	SET 7 BABUSHKA NESTING BOXES	7.65	01-12-10 8:26
536365	17850	GLASS STAR FROSTED T-LIGHT HOLDER	4.25	01-12-10 8:26
536366	17850	HAND WARMER UNION JACK	1.85	01-12-10 8:28
536366	17850	HAND WARMER RED POLKA DOT	1.85	01-12-10 8:28
536368	13047	JAM MAKING SET WITH JARS	4.25	01-12-10 8:34
536368	13047	RED COAT RACK PARIS FASHION	4.95	01-12-10 8:34
536368	13047	YELLOW COAT RACK PARIS FASHION	4.95	01-12-10 8:34
536368	13047	BLUE COAT RACK PARIS FASHION	4.95	01-12-10 8:34
536367	13047	ASSORTED COLOUR BIRD ORNAMENT	1.69	01-12-10 8:34
536367	13047	POPPY'S PLAYHOUSE BEDROOM	2.1	01-12-10 8:34
536367	13047	POPPY'S PLAYHOUSE KITCHEN	2.1	01-12-10 8:34
536367	13047	FELTCRAFT PRINCESS CHARLOTTE DOLL	3.75	01-12-10 8:34
536367	13047	IVORY KNITTED MUG COSY	1.65	01-12-10 8:34
536367	13047	BOX OF 6 ASSORTED COLOUR TEASPOONS	4.25	01-12-10 8:34
536367	13047	BOX OF VINTAGE JIGSAW BLOCKS	4.95	01-12-10 8:34
536367	13047	BOX OF VINTAGE ALPHABET BLOCKS	9.95	01-12-10 8:34
536367	13047	HOME BUILDING BLOCK WORD	5.95	01-12-10 8:34
536367	13047	LOVE BUILDING BLOCK WORD	5.95	01-12-10 8:34
536367	13047	RECIPE BOX WITH METAL HEART	7.95	01-12-10 8:34

Appendix 7: Creation of Cleaned Dataset

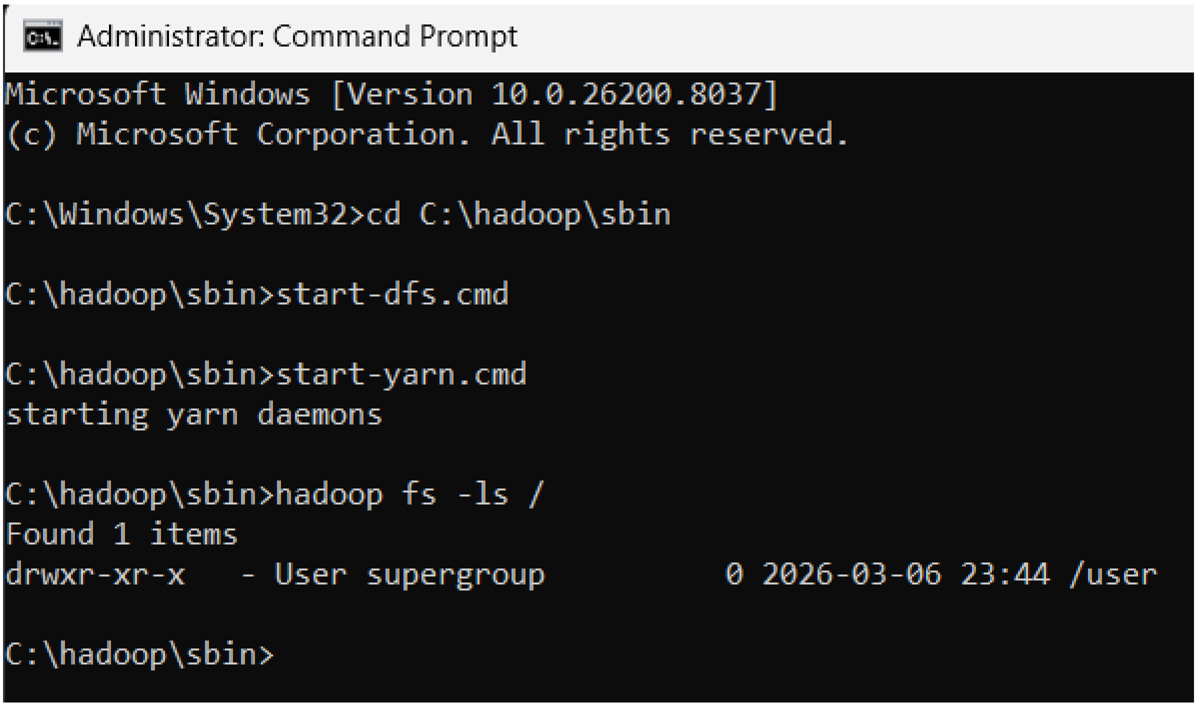


Appendix 8: Final Cleaned Dataset Saved

Part 2:- Hadoop Setup



Appendix 9: Hadoop Environment Initialization Command



Appendix 10: Hadoop Services Successfully Started

Part 3:- HDFS Data Upload

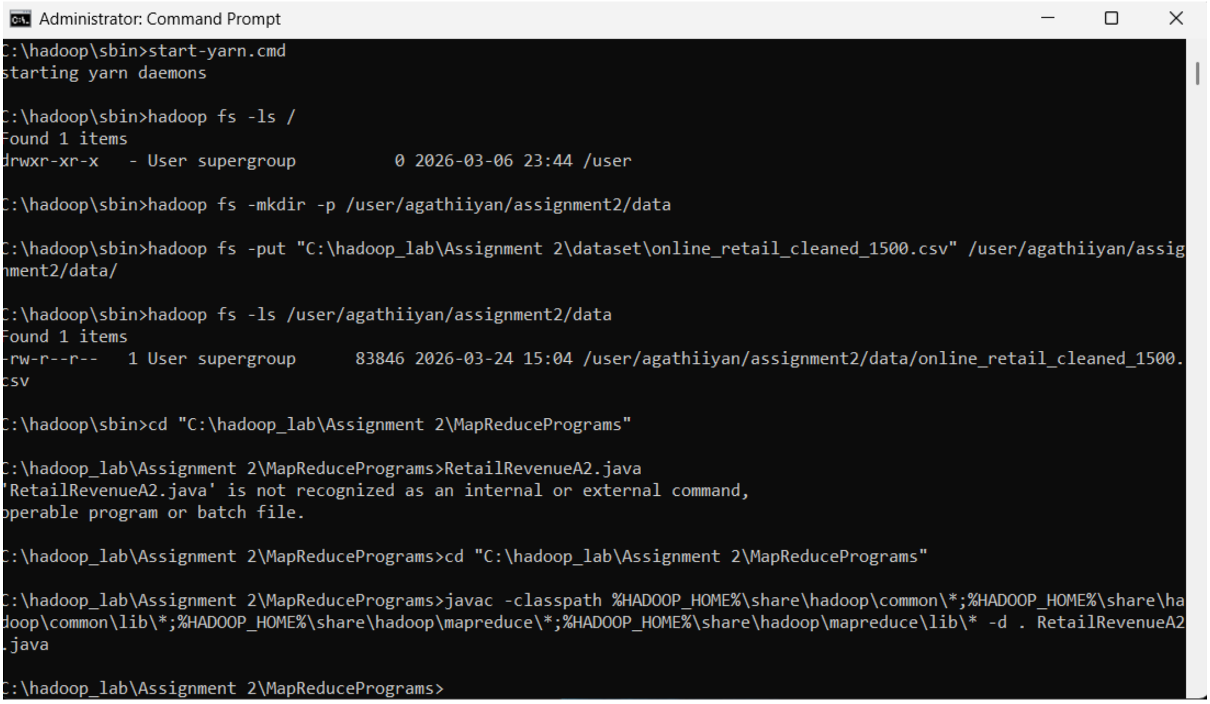
```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>cd C:\hadoop\sbin

C:\hadoop\sbin>start-dfs.cmd

C:\hadoop\sbin>start-yarn.cmd
starting yarn daemons

C:\hadoop\sbin>hadoop fs -ls /
Found 1 items
drwxr-xr-x   - User supergroup          0 2026-03-06 23:44 /user

C:\hadoop\sbin>hadoop fs -mkdir -p /user/agathiiyan/assignment2/data

C:\hadoop\sbin>hadoop fs -put "C:\hadoop_lab\Assignment 2\dataset\online_retail_cleaned_1500.csv" /user/agathiiyan/assig
nment2/data/

C:\hadoop\sbin>hadoop fs -ls /user/agathiiyan/assignment2/data
Found 1 items
-rw-r--r--   1 User supergroup      83846 2026-03-24 15:04 /user/agathiiyan/assignment2/data/online_retail_cleaned_1500.
csv

C:\hadoop\sbin>
```

Appendix 11: Uploading Dataset to HDFS

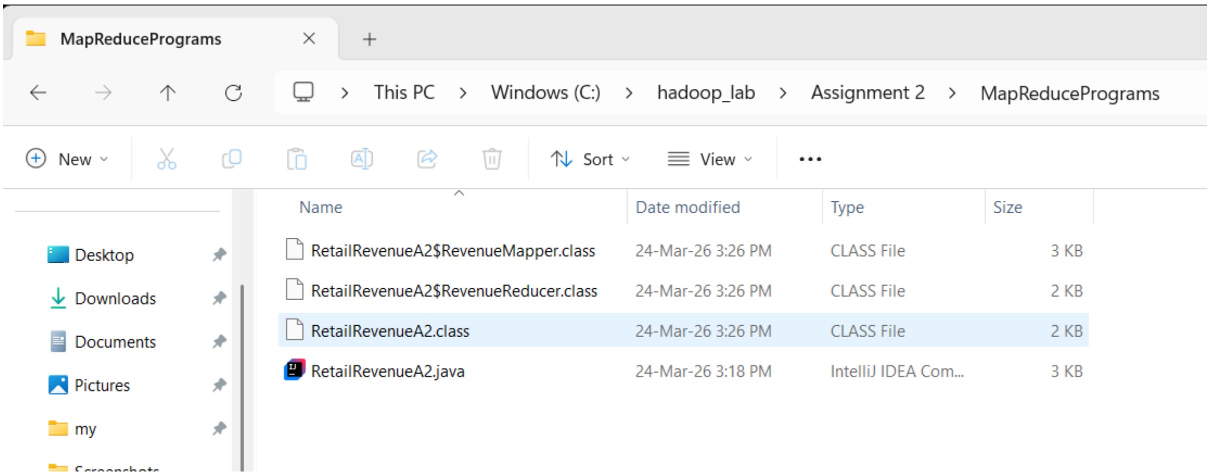
```
Administrator: Command Prompt
                Reduce output records=0
                Spilled Records=0
                Shuffled Maps =1
                Failed Shuffles=0
                Merged Map outputs=1
                GC time elapsed (ms)=4
                Total committed heap usage (bytes)=339738624
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=83846
        File Output Format Counters
                Bytes Written=0

C:\hadoop_lab\Assignment 2\MapReducePrograms>hadoop fs -ls /user/agathiiyan/assignment2/output_retail
Found 2 items
-rw-r--r--   1 User supergroup          0 2026-03-24 15:48 /user/agathiiyan/assignment2/output_retail/_SUCCESS
-rw-r--r--   1 User supergroup          0 2026-03-24 15:48 /user/agathiiyan/assignment2/output_retail/part-r-00000

C:\hadoop_lab\Assignment 2\MapReducePrograms>
```

Appendix 12: Verification of HDFS Directory Structure

Part 4:- MapReduce Development

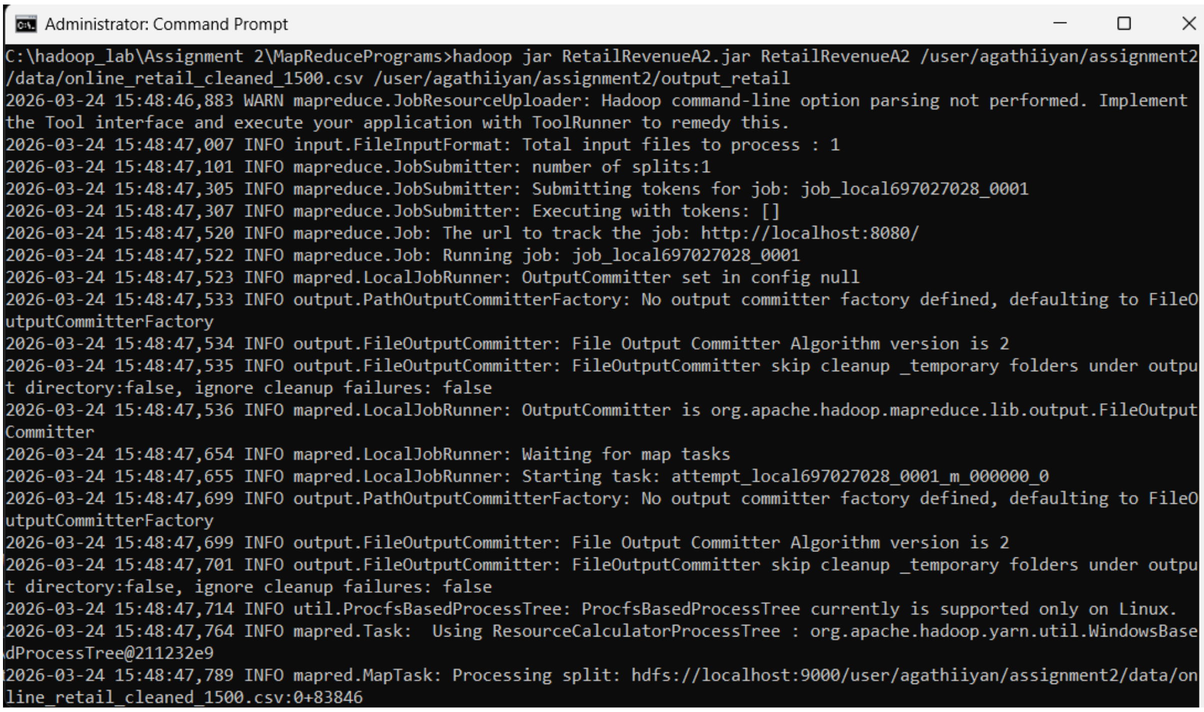


Appendix 13: Compilation of Java MapReduce Program



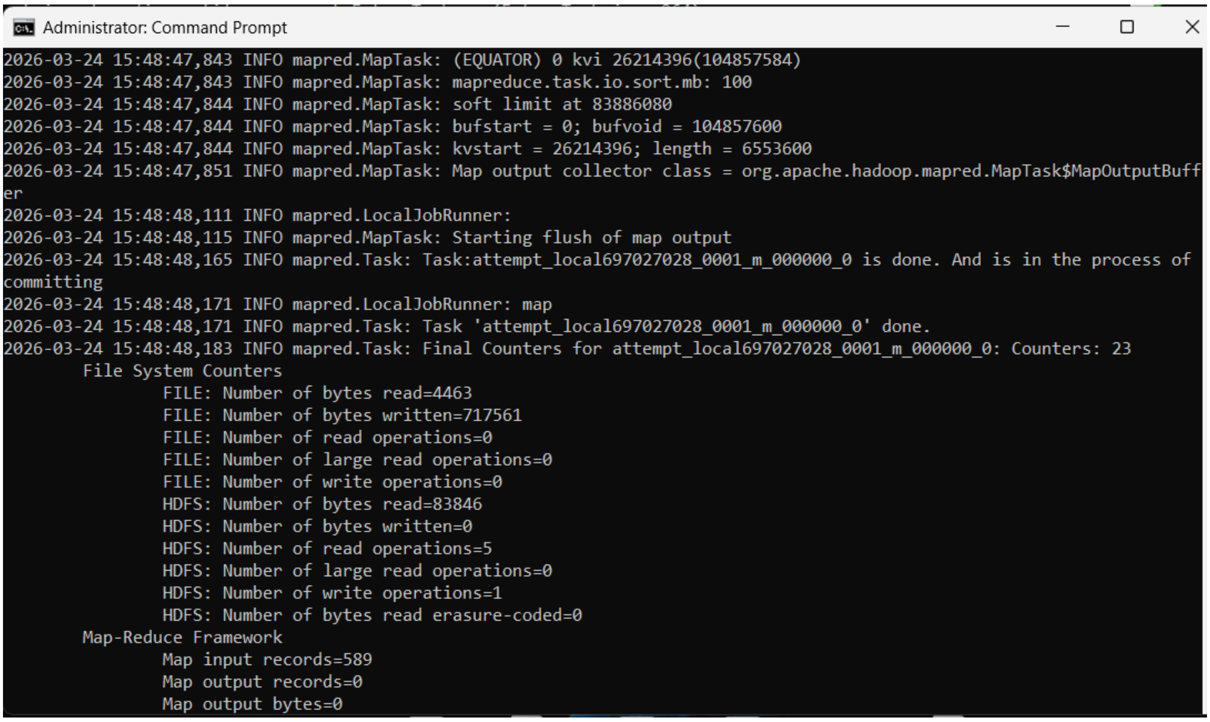
Appendix 14: Generated Class Files

Part 5:- MapReduce Execution



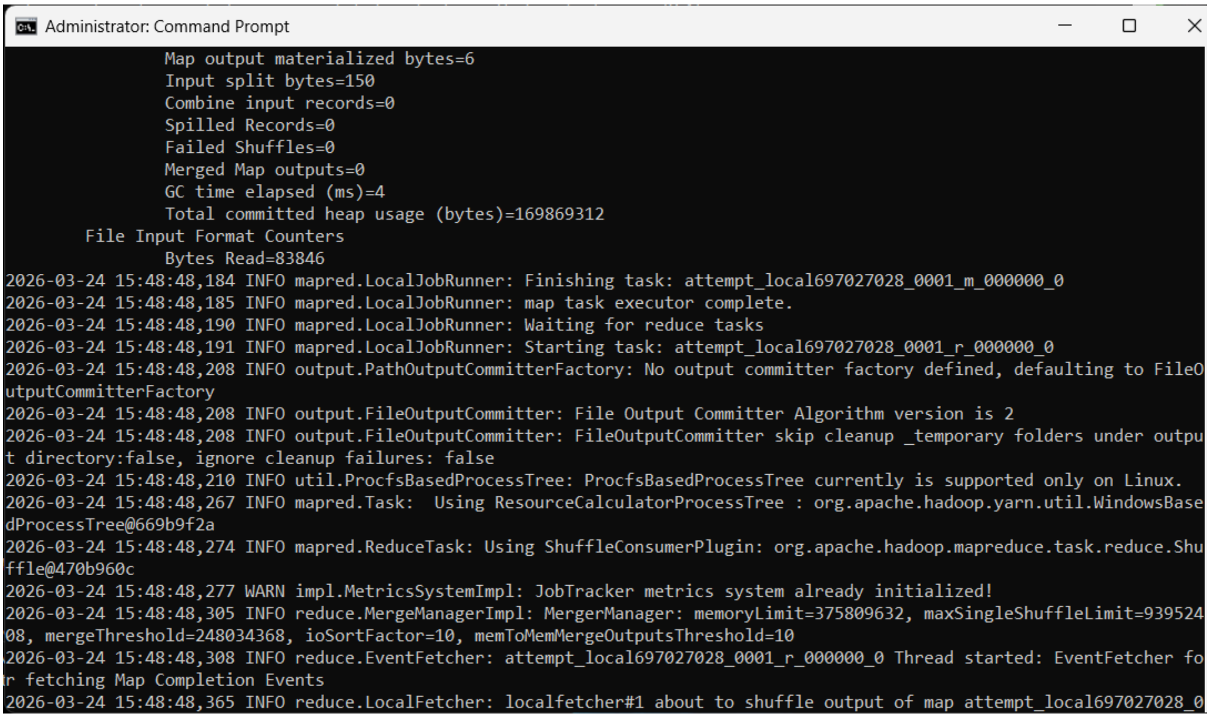
```
Administrator: Command Prompt
C:\hadoop_lab\Assignment 2\MapReducePrograms>hadoop jar RetailRevenueA2.jar RetailRevenueA2 /user/agathiiyan/assignment2
/data/online_retail_cleaned_1500.csv /user/agathiiyan/assignment2/output_retail
2026-03-24 15:48:46,883 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement
the Tool interface and execute your application with ToolRunner to remedy this.
2026-03-24 15:48:47,007 INFO input.FileInputFormat: Total input files to process : 1
2026-03-24 15:48:47,101 INFO mapreduce.JobSubmitter: number of splits:1
2026-03-24 15:48:47,305 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_local697027028_0001
2026-03-24 15:48:47,307 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-03-24 15:48:47,520 INFO mapreduce.Job: The url to track the job: http://localhost:8080/
2026-03-24 15:48:47,522 INFO mapreduce.Job: Running job: job_local697027028_0001
2026-03-24 15:48:47,523 INFO mapred.LocalJobRunner: OutputCommitter set in config null
2026-03-24 15:48:47,533 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileO
utputCommitterFactory
2026-03-24 15:48:47,534 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-03-24 15:48:47,535 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under outpu
t directory:false, ignore cleanup failures: false
2026-03-24 15:48:47,536 INFO mapred.LocalJobRunner: OutputCommitter is org.apache.hadoop.mapreduce.lib.output.FileOutput
Committer
2026-03-24 15:48:47,654 INFO mapred.LocalJobRunner: Waiting for map tasks
2026-03-24 15:48:47,655 INFO mapred.LocalJobRunner: Starting task: attempt_local697027028_0001_m_000000_0
2026-03-24 15:48:47,699 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileO
utputCommitterFactory
2026-03-24 15:48:47,699 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-03-24 15:48:47,701 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under outpu
t directory:false, ignore cleanup failures: false
2026-03-24 15:48:47,714 INFO util.ProcfsBasedProcessTree: ProcfsBasedProcessTree currently is supported only on Linux.
2026-03-24 15:48:47,764 INFO mapred.Task:  Using ResourceCalculatorProcessTree : org.apache.hadoop.yarn.util.WindowsBase
dProcessTree@211232e9
2026-03-24 15:48:47,789 INFO mapred.MapTask: Processing split: hdfs://localhost:9000/user/agathiiyan/assignment2/data/on
line_retail_cleaned_1500.csv:0+83846
```

Appendix 15: Execution of MapReduce Job



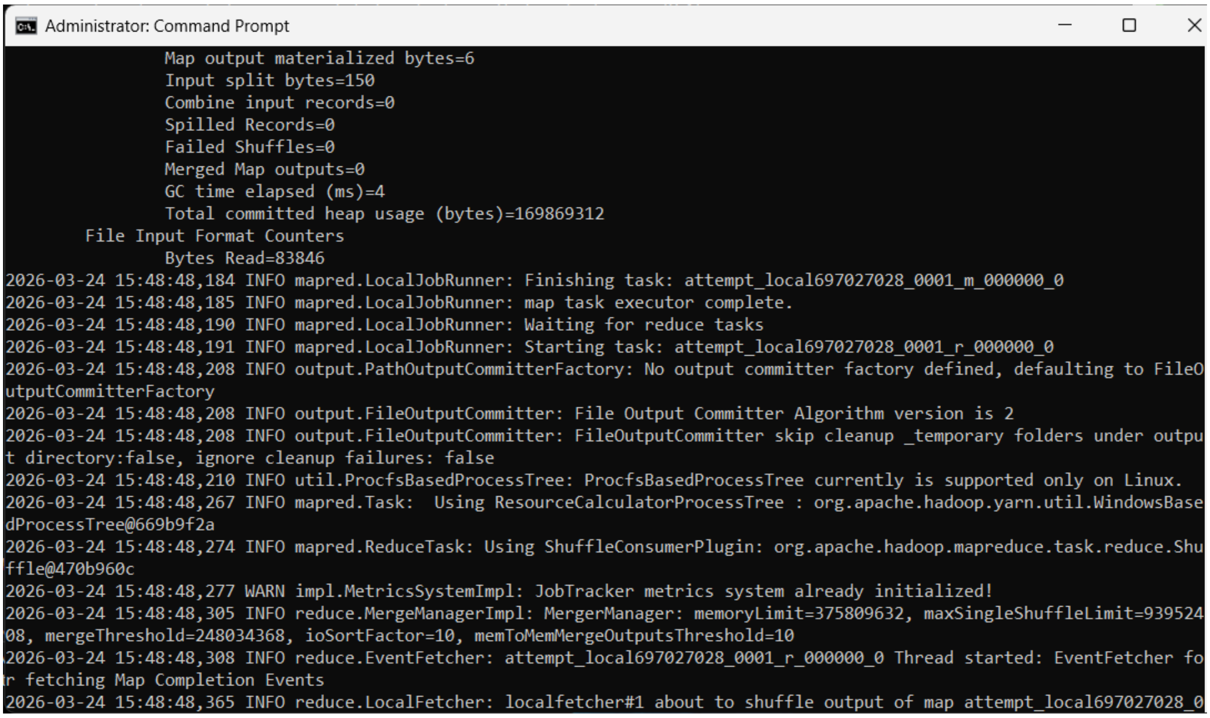
```
Administrator: Command Prompt
2026-03-24 15:48:47,843 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)
2026-03-24 15:48:47,843 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2026-03-24 15:48:47,844 INFO mapred.MapTask: soft limit at 83886080
2026-03-24 15:48:47,844 INFO mapred.MapTask: bufstart = 0; bufvoid = 104857600
2026-03-24 15:48:47,844 INFO mapred.MapTask: kvstart = 26214396; length = 6553600
2026-03-24 15:48:47,851 INFO mapred.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask$MapOutputBuff
er
2026-03-24 15:48:48,111 INFO mapred.LocalJobRunner:
2026-03-24 15:48:48,115 INFO mapred.MapTask: Starting flush of map output
2026-03-24 15:48:48,165 INFO mapred.Task: Task:attempt_local697027028_0001_m_000000_0 is done. And is in the process of
committing
2026-03-24 15:48:48,171 INFO mapred.LocalJobRunner: map
2026-03-24 15:48:48,171 INFO mapred.Task: Task 'attempt_local697027028_0001_m_000000_0' done.
2026-03-24 15:48:48,183 INFO mapred.Task: Final Counters for attempt_local697027028_0001_m_000000_0: Counters: 23
        File System Counters
                FILE: Number of bytes read=4463
                FILE: Number of bytes written=717561
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=83846
                HDFS: Number of bytes written=0
                HDFS: Number of read operations=5
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=1
                HDFS: Number of bytes read erasure-coded=0
        Map-Reduce Framework
                Map input records=589
                Map output records=0
                Map output bytes=0
```

Appendix 16: Execution of MapReduce Job



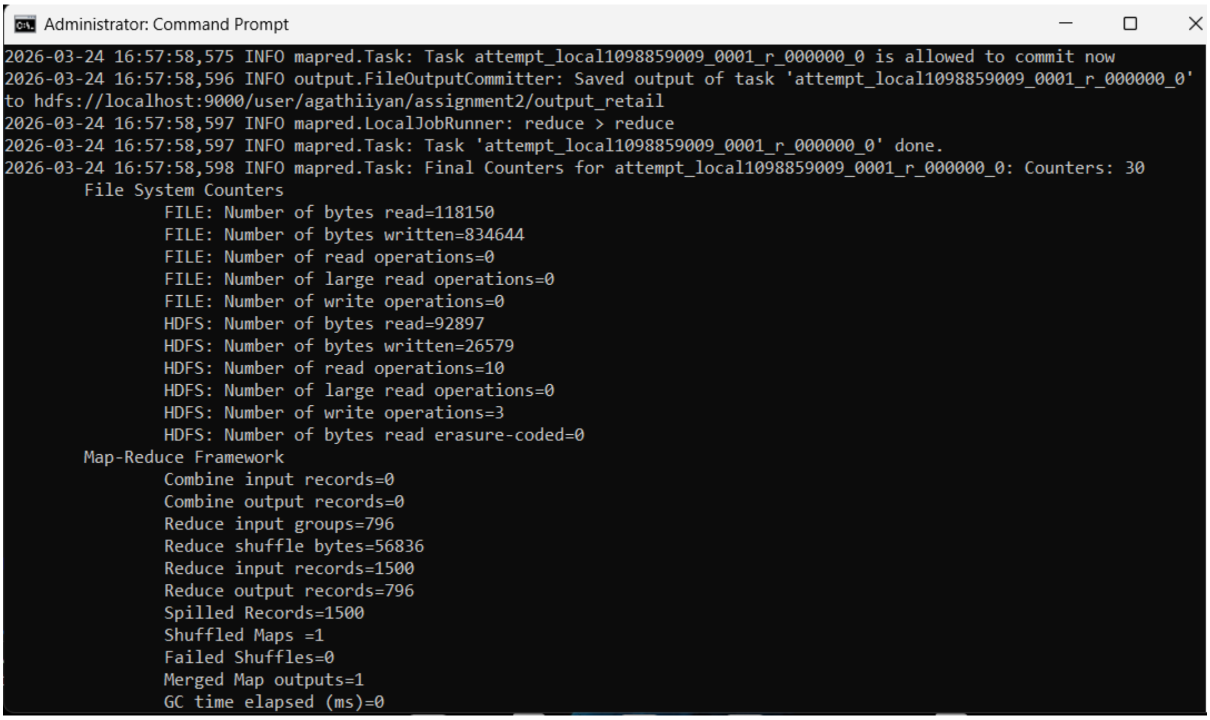
```
Administrator: Command Prompt
                Map output materialized bytes=6
                Input split bytes=150
                Combine input records=0
                Spilled Records=0
                Failed Shuffles=0
                Merged Map outputs=0
                GC time elapsed (ms)=4
                Total committed heap usage (bytes)=169869312
        File Input Format Counters
                Bytes Read=83846
2026-03-24 15:48:48,184 INFO mapred.LocalJobRunner: Finishing task: attempt_local697027028_0001_m_000000_0
2026-03-24 15:48:48,185 INFO mapred.LocalJobRunner: map task executor complete.
2026-03-24 15:48:48,190 INFO mapred.LocalJobRunner: Waiting for reduce tasks
2026-03-24 15:48:48,191 INFO mapred.LocalJobRunner: Starting task: attempt_local697027028_0001_r_000000_0
2026-03-24 15:48:48,208 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileO
utputCommitterFactory
2026-03-24 15:48:48,208 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-03-24 15:48:48,208 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under outpu
t directory:false, ignore cleanup failures: false
2026-03-24 15:48:48,210 INFO util.ProcfsBasedProcessTree: ProcfsBasedProcessTree currently is supported only on Linux.
2026-03-24 15:48:48,267 INFO mapred.Task:  Using ResourceCalculatorProcessTree : org.apache.hadoop.yarn.util.WindowsBase
dProcessTree@669b9f2a
2026-03-24 15:48:48,274 INFO mapred.ReduceTask: Using ShuffleConsumerPlugin: org.apache.hadoop.mapreduce.task.reduce.Shu
ffle@470b960c
2026-03-24 15:48:48,277 WARN impl.MetricsSystemImpl: JobTracker metrics system already initialized!
2026-03-24 15:48:48,305 INFO reduce.MergeManagerImpl: MergerManager: memoryLimit=375809632, maxSingleShuffleLimit=939524
08, mergeThreshold=248034368, ioSortFactor=10, memToMemMergeOutputsThreshold=10
2026-03-24 15:48:48,308 INFO reduce.EventFetcher: attempt_local697027028_0001_r_000000_0 Thread started: EventFetcher fo
r fetching Map Completion Events
2026-03-24 15:48:48,365 INFO reduce.LocalFetcher: localfetcher#1 about to shuffle output of map attempt_local697027028_0
```

Appendix 17: Execution of MapReduce Job



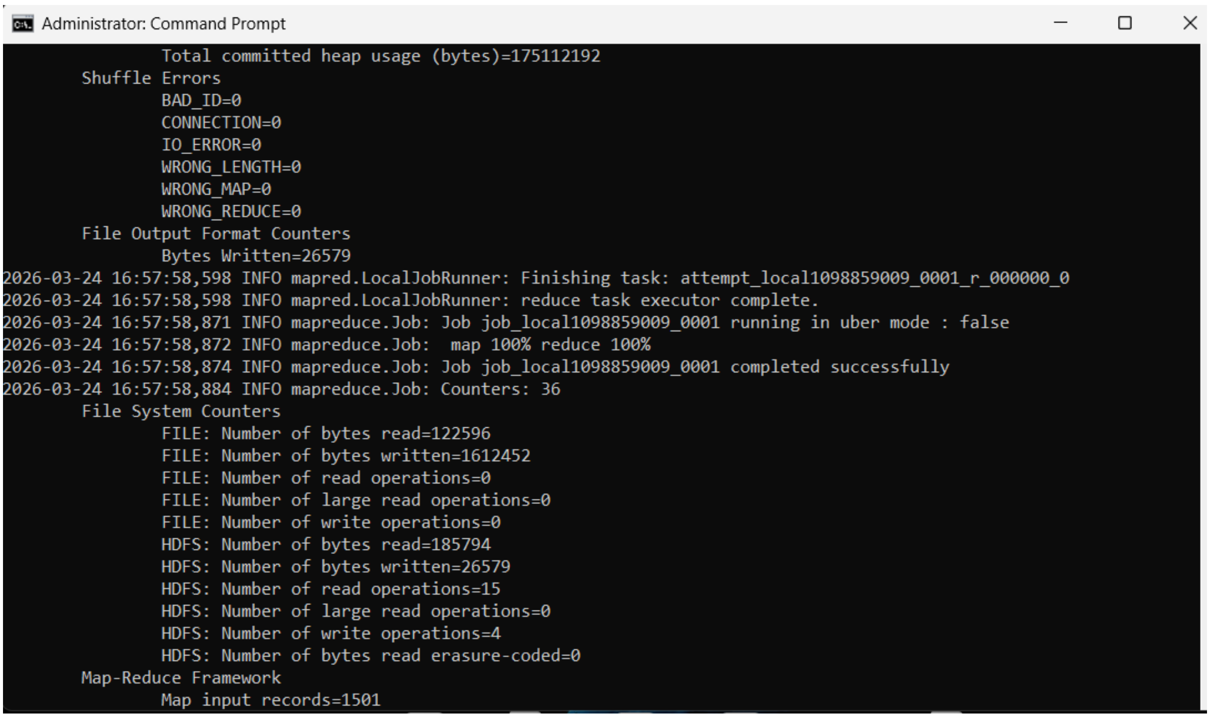
```
Administrator: Command Prompt
                Map output materialized bytes=6
                Input split bytes=150
                Combine input records=0
                Spilled Records=0
                Failed Shuffles=0
                Merged Map outputs=0
                GC time elapsed (ms)=4
                Total committed heap usage (bytes)=169869312
        File Input Format Counters
                Bytes Read=83846
2026-03-24 15:48:48,184 INFO mapred.LocalJobRunner: Finishing task: attempt_local697027028_0001_m_000000_0
2026-03-24 15:48:48,185 INFO mapred.LocalJobRunner: map task executor complete.
2026-03-24 15:48:48,190 INFO mapred.LocalJobRunner: Waiting for reduce tasks
2026-03-24 15:48:48,191 INFO mapred.LocalJobRunner: Starting task: attempt_local697027028_0001_r_000000_0
2026-03-24 15:48:48,208 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileO
utputCommitterFactory
2026-03-24 15:48:48,208 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-03-24 15:48:48,208 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under outpu
t directory:false, ignore cleanup failures: false
2026-03-24 15:48:48,210 INFO util.ProcfsBasedProcessTree: ProcfsBasedProcessTree currently is supported only on Linux.
2026-03-24 15:48:48,267 INFO mapred.Task:  Using ResourceCalculatorProcessTree : org.apache.hadoop.yarn.util.WindowsBase
dProcessTree@669b9f2a
2026-03-24 15:48:48,274 INFO mapred.ReduceTask: Using ShuffleConsumerPlugin: org.apache.hadoop.mapreduce.task.reduce.Shu
ffle@470b960c
2026-03-24 15:48:48,277 WARN impl.MetricsSystemImpl: JobTracker metrics system already initialized!
2026-03-24 15:48:48,305 INFO reduce.MergeManagerImpl: MergerManager: memoryLimit=375809632, maxSingleShuffleLimit=939524
08, mergeThreshold=248034368, ioSortFactor=10, memToMemMergeOutputsThreshold=10
2026-03-24 15:48:48,308 INFO reduce.EventFetcher: attempt_local697027028_0001_r_000000_0 Thread started: EventFetcher fo
r fetching Map Completion Events
2026-03-24 15:48:48,365 INFO reduce.LocalFetcher: localfetcher#1 about to shuffle output of map attempt_local697027028_0
```

Appendix 18: Execution of MapReduce Job



```
Administrator: Command Prompt
2026-03-24 16:57:58,575 INFO mapred.Task: Task attempt_local1098859009_0001_r_000000_0 is allowed to commit now
2026-03-24 16:57:58,596 INFO output.FileOutputCommitter: Saved output of task 'attempt_local1098859009_0001_r_000000_0' to hdfs://localhost:9000/user/agathiiyan/assignment2/output_retail
2026-03-24 16:57:58,597 INFO mapred.LocalJobRunner: reduce > reduce
2026-03-24 16:57:58,597 INFO mapred.Task: Task 'attempt_local1098859009_0001_r_000000_0' done.
2026-03-24 16:57:58,598 INFO mapred.Task: Final Counters for attempt_local1098859009_0001_r_000000_0: Counters: 30
        File System Counters
                FILE: Number of bytes read=118150
                FILE: Number of bytes written=834644
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=92897
                HDFS: Number of bytes written=26579
                HDFS: Number of read operations=10
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=3
                HDFS: Number of bytes read erasure-coded=0
        Map-Reduce Framework
                Combine input records=0
                Combine output records=0
                Reduce input groups=796
                Reduce shuffle bytes=56836
                Reduce input records=1500
                Reduce output records=796
                Spilled Records=1500
                Shuffled Maps =1
                Failed Shuffles=0
                Merged Map outputs=1
                GC time elapsed (ms)=0
```

Appendix 19: Execution of MapReduce Job



```
Administrator: Command Prompt
                Total committed heap usage (bytes)=175112192
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Output Format Counters
                Bytes Written=26579
2026-03-24 16:57:58,598 INFO mapred.LocalJobRunner: Finishing task: attempt_local1098859009_0001_r_000000_0
2026-03-24 16:57:58,598 INFO mapred.LocalJobRunner: reduce task executor complete.
2026-03-24 16:57:58,871 INFO mapreduce.Job: Job job_local1098859009_0001 running in uber mode : false
2026-03-24 16:57:58,872 INFO mapreduce.Job:  map 100% reduce 100%
2026-03-24 16:57:58,874 INFO mapreduce.Job: Job job_local1098859009_0001 completed successfully
2026-03-24 16:57:58,884 INFO mapreduce.Job: Counters: 36
        File System Counters
                FILE: Number of bytes read=122596
                FILE: Number of bytes written=1612452
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=185794
                HDFS: Number of bytes written=26579
                HDFS: Number of read operations=15
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=4
                HDFS: Number of bytes read erasure-coded=0
        Map-Reduce Framework
                Map input records=1501
```

Appendix 20: Execution of MapReduce Job

```
Administrator: Command Prompt
                Map output records=1500
                Map output bytes=53830
                Map output materialized bytes=56836
                Input split bytes=146
                Combine input records=0
                Combine output records=0
                Reduce input groups=796
                Reduce shuffle bytes=56836
                Reduce input records=1500
                Reduce output records=796
                Spilled Records=3000
                Shuffled Maps =1
                Failed Shuffles=0
                Merged Map outputs=1
                GC time elapsed (ms)=4
                Total committed heap usage (bytes)=350224384
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=92897
        File Output Format Counters
                Bytes Written=26579

C:\hadoop_lab\Assignment 2\MapReducePrograms>
```

Appendix 21: Execution of MapReduce Job

Part 6:- Output Result

```
Administrator: Command Prompt
                Combine input records=0
                Combine output records=0
                Reduce input groups=796
                Reduce shuffle bytes=56836
                Reduce input records=1500
                Reduce output records=796
                Spilled Records=3000
                Shuffled Maps =1
                Failed Shuffles=0
                Merged Map outputs=1
                GC time elapsed (ms)=4
                Total committed heap usage (bytes)=350224384
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=92897
        File Output Format Counters
                Bytes Written=26579

C:\hadoop_lab\Assignment 2\MapReducePrograms>hadoop fs -ls /user/agathiiyan/assignment2/output_retail
Found 2 items
-rw-r--r--   1 User supergroup          0 2026-03-24 16:57 /user/agathiiyan/assignment2/output_retail/_SUCCESS
-rw-r--r--   1 User supergroup      26579 2026-03-24 16:57 /user/agathiiyan/assignment2/output_retail/part-r-00000

C:\hadoop_lab\Assignment 2\MapReducePrograms>
```

Appendix 22: Output Directory after Job Execution

Administrator: Command Prompt

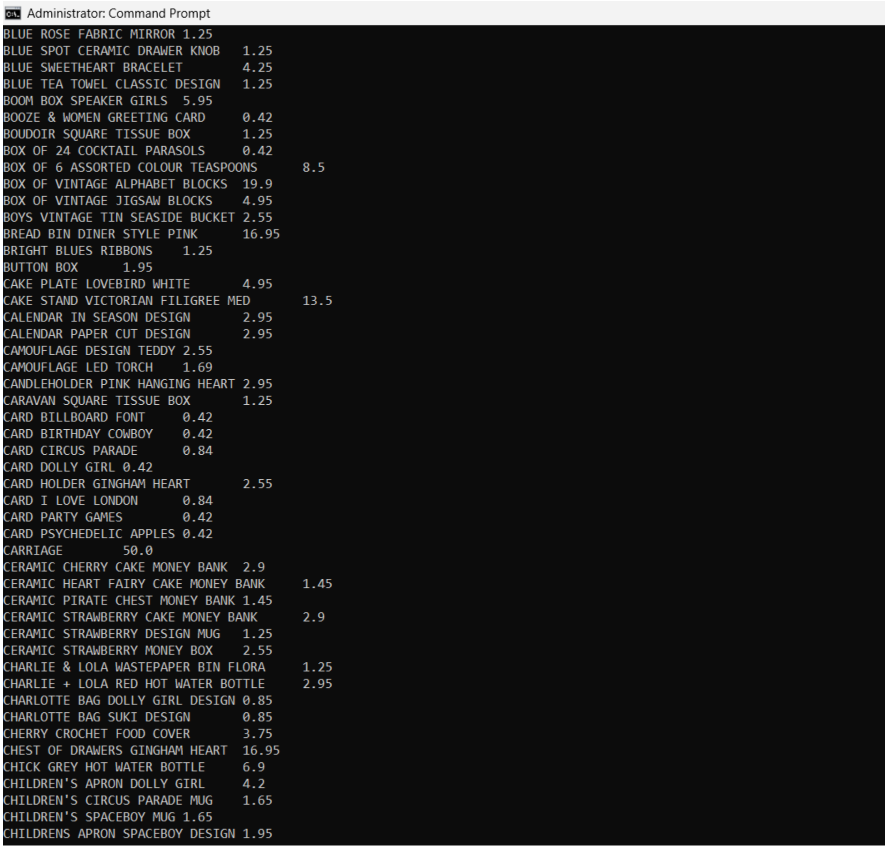
```
C:\hadoop_lab\Assignment 2\MapReducePrograms>hadoop fs -cat /user/agathiiyan/assignment2/output_retail/part-r-00000
"CHARLIE+LOLA""EXTREMELY BUSY"" SIGN"   2.55
"RECORD FRAME 7"" SINGLE SIZE " 4.65
10 COLOUR SPACEBOY PEN  1.7
12 DAISY PEGS IN WOOD BOX       8.25
12 MESSAGE CARDS WITH ENVELOPES 1.65
12 PENCIL SMALL TUBE WOODLAND   0.65
12 PENCILS SMALL TUBE SKULL     0.65
12 PENCILS TALL TUBE SKULLS     0.85
20 DOLLY PEGS RETROSPOT 1.25
200 RED + WHITE BENDY STRAWS    1.25
3 HOOK HANGER MAGIC GARDEN      1.95
3 HOOK PHOTO SHELF ANTIQUE WHITE        8.5
3 PIECE SPACEBOY COOKIE CUTTER SET      4.2
3 STRIPEY MICE FELTCRAFT        5.85
3 TIER CAKE TIN GREEN AND CREAM 44.849999999999994
3 TIER CAKE TIN RED AND CREAM   14.95
36 FOIL HEART CAKE CASES        6.300000000000001
36 FOIL STAR CAKE CASES 2.1
36 PENCILS TUBE RED RETROSPOT   1.25
36 PENCILS TUBE WOODLAND        1.25
3D CHRISTMAS STAMPS STICKERS    2.5
3D DOG PICTURE PLAYING CARDS    2.95
4 IVORY DINNER CANDLES SILVER FLOCK     2.55
4 PURPLE FLOCK DINNER CANDLES   2.55
4 TRADITIONAL SPINNING TOPS     2.5
5 HOOK HANGER MAGIC TOADSTOOL   4.949999999999999
5 HOOK HANGER RED MAGIC TOADSTOOL       6.6
5 STRAND GLASS NECKLACE CRYSTAL 19.049999999999997
6 RIBBONS ELEGANT CHRISTMAS     1.65
6 RIBBONS RUSTIC CHARM  6.6
6 RIBBONS SHIMMERING PINKS      1.65
60 CAKE CASES DOLLY GIRL DESIGN 0.55
60 CAKE CASES VINTAGE CHRISTMAS 1.1
60 TEATIME FAIRY CAKE CASES     2.2
72 SWEETHEART FAIRY CAKE CASES  1.6500000000000001
ADVENT CALENDAR GINGHAM SACK    5.95
AGED GLASS SILVER T-LIGHT HOLDER        2.5
AIRLINE BAG VINTAGE JET SET WHITE       4.25
AIRLINE BAG VINTAGE TOKYO 78    4.25
AIRLINE BAG VINTAGE WORLD CHAMPION      4.25
AIRLINE LOUNGE-METAL SIGN       2.1
ALARM CLOCK BAKELIKE GREEN      18.75
ALARM CLOCK BAKELIKE IVORY      11.25
ALARM CLOCK BAKELIKE ORANGE     3.75
ALARM CLOCK BAKELIKE PINK       3.75
ALARM CLOCK BAKELIKE RED        15.0
ANGEL DECORATION STARS ON DRESS 0.42
ANTIQUE GLASS DRESSING TABLE POT        5.9
```

Appendix 23: MapReduce Output

Administrator: Command Prompt

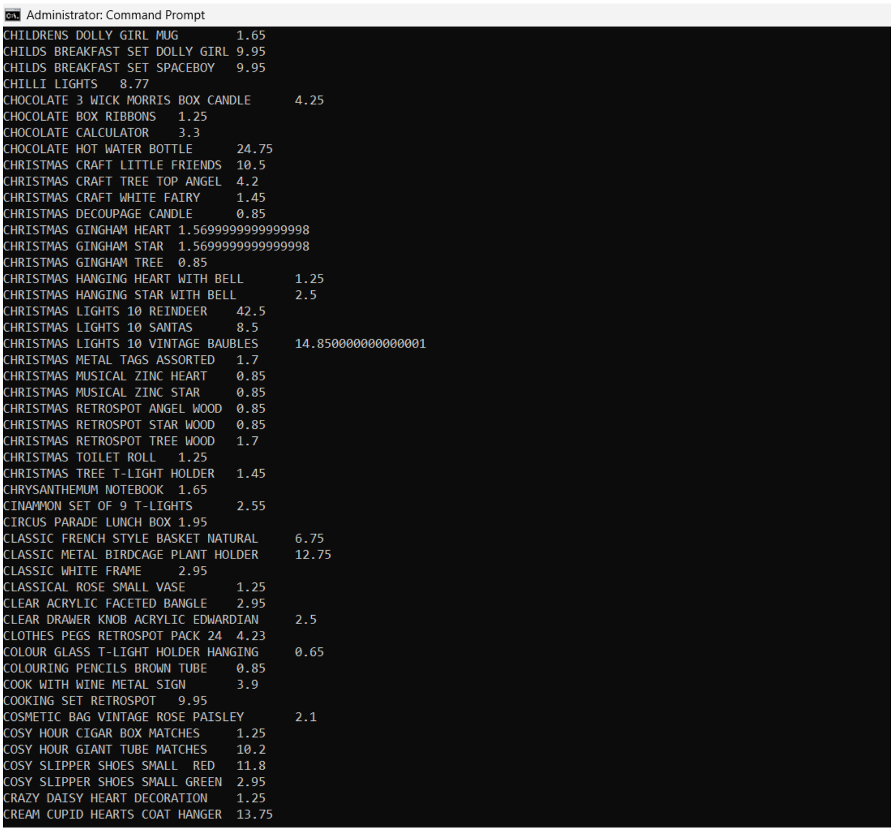
```
ANTIQUE GLASS HEART DECORATION  1.95
ANTIQUE GLASS PEDESTAL BOWL     3.75
ANTIQUE SILVER TEA GLASS ENGRAVED       1.06
ANTIQUE TALL SWIRLGLASS TRINKET POT     3.75
APPLE BATH SPONGE       1.25
ASS COL SMALL SAND GECKO P'WEIGHT       0.42
ASSORTED BOTTLE TOP  MAGNETS    0.42
ASSORTED COLOUR BIRD ORNAMENT   13.519999999999998
ASSORTED COLOUR LIZARD SUCTION HOOK     0.42
ASSORTED COLOUR MINI CASES      23.85
BABUSHKA LIGHTS STRING OF 10    27.0
BAG 500g SWIRLY MARBLES 1.65
BALLOON ART MAKE YOUR OWN FLOWERS       1.95
BALLOONS  WRITING SET   3.3
BATH BUILDING BLOCK WORD        5.95
BATHROOM METAL SIGN     0.55
BIRD DECORATION RED RETROSPOT   0.85
BIRD HOUSE HOT WATER BOTTLE     5.1
BIRTHDAY CARD- RETRO SPOT       0.42
BIRTHDAY PARTY CORDON BARRIER TAPE      1.25
BISCUIT TIN VINTAGE RED 6.75
BLACK CANDELABRA T-LIGHT HOLDER 4.2
BLACK DIAMANTE EXPANDABLE RING  4.25
BLACK DINER WALL CLOCK  8.5
BLACK HEART CARD HOLDER 7.34
BLACK KITCHEN SCALES    8.5
BLACK LOVE BIRD CANDLE  2.5
BLACK ORANGE SQUEEZER   0.42
BLACK PIRATE TREASURE CHEST     1.65
BLACK RECORD COVER FRAME        3.39
BLACK SWEETHEART BRACELET       4.25
BLACK TEA TOWEL CLASSIC DESIGN  2.5
BLACK/BLUE POLKADOT UMBRELLA    17.85
BLUE 3 PIECE POLKADOT CUTLERY SET       7.5
BLUE BIRDHOUSE DECORATION       0.85
BLUE CHARLIE+LOLA PERSONAL DOORSIGN     2.95
BLUE COAT RACK PARIS FASHION    9.9
BLUE DINER WALL CLOCK   17.0
BLUE DRAGONFLY HELICOPTER       7.95
BLUE DRAWER KNOB ACRYLIC EDWARDIAN      1.25
BLUE GIANT GARDEN THERMOMETER   5.95
BLUE HARMONICA IN BOX   1.25
BLUE NEW BAROQUE CANDLESTICK CANDLE     8.850000000000001
BLUE OWL SOFT TOY       2.95
BLUE PAISLEY NOTEBOOK   1.65
BLUE PAISLEY TISSUE BOX 2.5
BLUE PAPER PARASOL      2.95
BLUE PARTY BAGS 2.1
BLUE POLKADOT PLATE     1.69
```

Appendix 24: MapReduce Output



```
Administrator: Command Prompt
BLUE ROSE FABRIC MIRROR 1.25
BLUE SPOT CERAMIC DRAWER KNOB   1.25
BLUE SWEETHEART BRACELET        4.25
BLUE TEA TOWEL CLASSIC DESIGN   1.25
BOOM BOX SPEAKER GIRLS  5.95
BOOZE & WOMEN GREETING CARD     0.42
BOUDOIR SQUARE TISSUE BOX       1.25
BOX OF 24 COCKTAIL PARASOLS     0.42
BOX OF 6 ASSORTED COLOUR TEASPOONS      8.5
BOX OF VINTAGE ALPHABET BLOCKS  19.9
BOX OF VINTAGE JIGSAW BLOCKS    4.95
BOYS VINTAGE TIN SEASIDE BUCKET 2.55
BREAD BIN DINER STYLE PINK      16.95
BRIGHT BLUES RIBBONS    1.25
BUTTON BOX      1.95
CAKE PLATE LOVEBIRD WHITE       4.95
CAKE STAND VICTORIAN FILIGREE MED       13.5
CALENDAR IN SEASON DESIGN       2.95
CALENDAR PAPER CUT DESIGN       2.95
CAMOUFLAGE DESIGN TEDDY 2.55
CAMOUFLAGE LED TORCH    1.69
CANDLEHOLDER PINK HANGING HEART 2.95
CARAVAN SQUARE TISSUE BOX       1.25
CARD BILLBOARD FONT     0.42
CARD BIRTHDAY COWBOY    0.42
CARD CIRCUS PARADE      0.84
CARD DOLLY GIRL 0.42
CARD HOLDER GINGHAM HEART       2.55
CARD I LOVE LONDON      0.84
CARD PARTY GAMES        0.42
CARD PSYCHEDELIC APPLES 0.42
CARRIAGE        50.0
CERAMIC CHERRY CAKE MONEY BANK  2.9
CERAMIC HEART FAIRY CAKE MONEY BANK     1.45
CERAMIC PIRATE CHEST MONEY BANK 1.45
CERAMIC STRAWBERRY CAKE MONEY BANK      2.9
CERAMIC STRAWBERRY DESIGN MUG   1.25
CERAMIC STRAWBERRY MONEY BOX    2.55
CHARLIE & LOLA WASTEPAPER BIN FLORA     1.25
CHARLIE + LOLA RED HOT WATER BOTTLE     2.95
CHARLOTTE BAG DOLLY GIRL DESIGN 0.85
CHARLOTTE BAG SUKI DESIGN       0.85
CHERRY CROCHET FOOD COVER       3.75
CHEST OF DRAWERS GINGHAM HEART  16.95
CHICK GREY HOT WATER BOTTLE     6.9
CHILDREN'S APRON DOLLY GIRL     4.2
CHILDREN'S CIRCUS PARADE MUG    1.65
CHILDREN'S SPACEBOY MUG 1.65
CHILDRENS APRON SPACEBOY DESIGN 1.95
```

Appendix 25: MapReduce Output



```
Administrator: Command Prompt
CHILDRENS DOLLY GIRL MUG        1.65
CHILDS BREAKFAST SET DOLLY GIRL 9.95
CHILDS BREAKFAST SET SPACEBOY   9.95
CHILLI LIGHTS   8.77
CHOCOLATE 3 WICK MORRIS BOX CANDLE      4.25
CHOCOLATE BOX RIBBONS   1.25
CHOCOLATE CALCULATOR    3.3
CHOCOLATE HOT WATER BOTTLE      24.75
CHRISTMAS CRAFT LITTLE FRIENDS  10.5
CHRISTMAS CRAFT TREE TOP ANGEL  4.2
CHRISTMAS CRAFT WHITE FAIRY     1.45
CHRISTMAS DECOUPAGE CANDLE      0.85
CHRISTMAS GINGHAM HEART 1.5699999999999998
CHRISTMAS GINGHAM STAR  1.5699999999999998
CHRISTMAS GINGHAM TREE  0.85
CHRISTMAS HANGING HEART WITH BELL       1.25
CHRISTMAS HANGING STAR WITH BELL        2.5
CHRISTMAS LIGHTS 10 REINDEER    42.5
CHRISTMAS LIGHTS 10 SANTAS      8.5
CHRISTMAS LIGHTS 10 VINTAGE BAUBLES     14.850000000000001
CHRISTMAS METAL TAGS ASSORTED   1.7
CHRISTMAS MUSICAL ZINC HEART    0.85
CHRISTMAS MUSICAL ZINC STAR     0.85
CHRISTMAS RETROSPOT ANGEL WOOD  0.85
CHRISTMAS RETROSPOT STAR WOOD   0.85
CHRISTMAS RETROSPOT TREE WOOD   1.7
CHRISTMAS TOILET ROLL   1.25
CHRISTMAS TREE T-LIGHT HOLDER   1.45
CHRYSANTHEMUM NOTEBOOK  1.65
CINAMMON SET OF 9 T-LIGHTS      2.55
CIRCUS PARADE LUNCH BOX 1.95
CLASSIC FRENCH STYLE BASKET NATURAL     6.75
CLASSIC METAL BIRDCAGE PLANT HOLDER     12.75
CLASSIC WHITE FRAME     2.95
CLASSICAL ROSE SMALL VASE       1.25
CLEAR ACRYLIC FACETED BANGLE    2.95
CLEAR DRAWER KNOB ACRYLIC EDWARDIAN     2.5
CLOTHES PEGS RETROSPOT PACK 24  4.23
COLOUR GLASS T-LIGHT HOLDER HANGING     0.65
COLOURING PENCILS BROWN TUBE    0.85
COOK WITH WINE METAL SIGN       3.9
COOKING SET RETROSPOT   9.95
COSMETIC BAG VINTAGE ROSE PAISLEY       2.1
COSY HOUR CIGAR BOX MATCHES     1.25
COSY HOUR GIANT TUBE MATCHES    10.2
COSY SLIPPER SHOES SMALL  RED   11.8
COSY SLIPPER SHOES SMALL GREEN  2.95
CRAZY DAISY HEART DECORATION    1.25
CREAM CUPID HEARTS COAT HANGER  13.75
```

Appendix 26: MapReduce Output

Administrator: Command Prompt

```
CREAM HEART CARD HOLDER 7.34
CREAM SLICE FLANNEL CHOCOLATE SPOT      2.95
CREAM SLICE FLANNEL PINK SPOT   2.95
CREAM SWEETHEART EGG HOLDER     4.95
CREAM SWEETHEART LETTER RACK    5.45
CREAM SWEETHEART WALL CABINET   18.95
CUPCAKE LACE PAPER SET 6        7.8
CURIOUS  IMAGES NOTEBOOK SET    4.25
CUSHION COVER PINK UNION JACK   5.95
DECORATIVE PLANT POT WITH FRIEZE        7.95
DELUXE SEWING KIT       4.95
DIAMANTE HAIR GRIP PACK/2 BLACK DIA     1.65
DIAMANTE HAIR GRIP PACK/2 MONTANA       1.65
DIAMANTE HAIR GRIP PACK/2 RUBY  1.65
DINOSAURS  WRITING SET  1.65
DISCO BALL CHRISTMAS DECORATION 0.33999999999999997
DISCO BALL ROTATOR BATTERY OPERATED     4.25
DOCTOR'S BAG SOFT TOY   8.95
DOG BOWL CHASING BALL DESIGN    3.75
DOLLY GIRL LUNCH BOX    3.9
DOORMAT BLACK FLOCK     7.95
DOORMAT ENGLISH ROSE    7.95
DOORMAT FAIRY CAKE      7.95
DOORMAT FANCY FONT HOME SWEET HOME      30.599999999999998
DOORMAT HEARTS  15.9
DOORMAT NEW ENGLAND     7.95
DOORMAT PEACE ON EARTH BLUE     7.95
DOORMAT RED RETROSPOT   23.85
DOORMAT SPOTTY HOME SWEET HOME  7.95
DOORMAT TOPIARY 23.85
DOORMAT UNION FLAG      7.95
DOORMAT UNION JACK GUNS AND ROSES       7.95
DOORSTOP RETROSPOT HEART        3.75
DOVE DECORATION PAINTED ZINC    0.65
EDWARDIAN PARASOL BLACK 16.85
EDWARDIAN PARASOL NATURAL       11.9
EDWARDIAN PARASOL RED   25.75
EIGHT PIECE DINOSAUR SET        1.25
EMERGENCY FIRST AID TIN 1.25
ENAMEL BREAD BIN CREAM  10.95
ENAMEL FIRE BUCKET CREAM        20.85
ENAMEL FLOWER JUG CREAM 5.95
ENAMEL MEASURING JUG CREAM      4.25
ENGLISH ROSE HOT WATER BOTTLE   12.25
ENGLISH ROSE SPIRIT LEVEL       1.95
FAIRY CAKE BIRTHDAY CANDLE SET  3.75
FAIRY CAKE FLANNEL ASSORTED COLOUR      2.55
FAIRY SOAP SOAP HOLDER  1.69
FAIRY TALE COTTAGE NIGHTLIGHT   3.4
```

Appendix 27: MapReduce Output

Administrator: Command Prompt

```
FAMILY PHOTO FRAME CORNICE      39.8
FANCY FONT BIRTHDAY CARD-       1.26
FANCY FONTS BIRTHDAY WRAP       0.42
FAWN BLUE HOT WATER BOTTLE      5.9
FELT EGG COSY CHICKEN   0.85
FELT EGG COSY WHITE RABBIT      0.85
FELTCRAFT 6 FLOWER FRIENDS      8.4
FELTCRAFT BUTTERFLY HEARTS      2.9
FELTCRAFT CHRISTMAS FAIRY       17.0
FELTCRAFT CUSHION BUTTERFLY     11.25
FELTCRAFT CUSHION OWL   15.0
FELTCRAFT CUSHION RABBIT        11.25
FELTCRAFT DOLL EMILY    2.95
FELTCRAFT DOLL MOLLY    8.850000000000001
FELTCRAFT DOLL ROSIE    2.95
FELTCRAFT HAIRBAND PINK AND PURPLE      1.7
FELTCRAFT HAIRBAND RED AND BLUE 0.85
FELTCRAFT HAIRBANDS PINK AND WHITE      0.85
FELTCRAFT PRINCESS CHARLOTTE DOLL       26.25
FELTCRAFT PRINCESS LOLA DOLL    3.75
FELTCRAFT PRINCESS OLIVIA DOLL  3.75
FIRST AID TIN   6.5
FIVE HEART HANGING DECORATION   2.95
FLOWERS  STICKERS       0.85
FOLDING UMBRELLA CREAM POLKADOT 9.9
FOLDING UMBRELLA PINKWHITE POLKADOT     4.95
FOLDING UMBRELLA RED/WHITE POLKADOT     9.9
FOLDING UMBRELLA WHITE/RED POLKADOT     4.95
FOLKART ZINC HEART CHRISTMAS DEC        0.85
FOUR HOOK  WHITE LOVEBIRDS      2.1
FRENCH BLUE METAL DOOR SIGN 5   1.25
FRENCH WC SIGN BLUE METAL       2.5
FRIDGE MAGNETS LES ENFANTS ASSORTED     0.85
FRIDGE MAGNETS US DINER ASSORTED        0.85
FULL ENGLISH BREAKFAST PLATE    3.75
GARDEN PATH JOURNAL     2.55
GARLAND WOODEN HAPPY EASTER     1.25
GENTLEMAN SHIRT REPAIR KIT      0.85
GIN & TONIC DIET GREETING CARD  0.42
GIN + TONIC DIET METAL SIGN     4.2
GIN AND TONIC MUG       1.25
GINGERBREAD MAN COOKIE CUTTER   1.25
GINGHAM HEART  DOORSTOP RED     8.5
GIRLS VINTAGE TIN SEASIDE BUCKET        2.55
GIRLY PINK TOOL SET     4.95
GLASS  BEURRE DISH      3.95
GLASS CLOCHE SMALL      3.95
GLASS JAR DIGESTIVE BISCUITS    2.95
GLASS JAR ENGLISH CONFECTIONERY 2.95
```

Appendix 28: MapReduce Output

```
Administrator: Command Prompt
GLASS STAR FROSTED T-LIGHT HOLDER       21.25
GRAND CHOCOLATECANDLE   1.45
GREEN CHRISTMAS TREE CARD HOLDER        4.25
GREEN DRAWER KNOB ACRYLIC EDWARDIAN     1.25
GREEN FERN NOTEBOOK     1.65
GREEN GIANT GARDEN THERMOMETER  5.95
GREEN POLKADOT PLATE    1.69
GREEN REGENCY TEACUP AND SAUCER 2.95
GREY HEART HOT WATER BOTTLE     15.0
GROW YOUR OWN BASIL IN ENAMEL MUG       6.300000000000001
GROW YOUR OWN PLANT IN A CAN    1.06
GUMBALL COAT RACK       2.55
GUMBALL MAGAZINE RACK   7.65
GUMBALL MONOCHROME COAT RACK    1.06
HAND OVER THE CHOCOLATE   SIGN  4.2
HAND WARMER BABUSHKA DESIGN     4.2
HAND WARMER BIRD DESIGN 14.45
HAND WARMER OWL DESIGN  20.750000000000004
HAND WARMER RED POLKA DOT       9.25
HAND WARMER RED RETROSPOT       20.750000000000004
HAND WARMER SCOTTY DOG DESIGN   27.050000000000008
HAND WARMER UNION JACK  19.5
HANGING GLASS ETCHED TEALIGHT   3.3
HANGING HEART MIRROR DECORATION 0.65
HANGING HEART ZINC T-LIGHT HOLDER       1.49
HANGING MEDINA LANTERN SMALL    5.9
HANGING METAL HEART LANTERN     3.3
HANGING METAL STAR LANTERN      1.65
HEADS AND TAILS SPORTING FUN    1.25
HEART FILIGREE DOVE  SMALL      5.0
HEART FILIGREE DOVE LARGE       1.65
HEART IVORY TRELLIS LARGE       1.65
HEART IVORY TRELLIS SMALL       2.5
HEART OF WICKER LARGE   11.399999999999999
HEART OF WICKER SMALL   11.350000000000001
HEART T-LIGHT HOLDER    2.9
HEART WOODEN CHRISTMAS DECORATION       0.85
HERB MARKER BASIL       0.65
HERB MARKER CHIVES      0.65
HERB MARKER MINT        0.65
HERB MARKER PARSLEY     0.65
HERB MARKER ROSEMARY    0.65
HERB MARKER THYME       0.65
HI TEC ALPINE HAND WARMER       3.3
HOME BUILDING BLOCK WORD        23.8
HOME SMALL WOOD LETTERS 4.95
HOMEMADE JAM SCENTED CANDLES    9.95
HOT WATER BOTTLE BABUSHKA       32.15
HOT WATER BOTTLE I AM SO POORLY 9.3
```

Appendix 29: MapReduce Output

```
Administrator: Command Prompt
HOT WATER BOTTLE TEA AND SYMPATHY       23.2
HYACINTH BULB T-LIGHT CANDLES   0.84
I CAN ONLY PLEASE ONE PERSON MUG        1.25
I'M ON HOLIDAY METAL SIGN       2.1
IF YOU CAN'T STAND THE HEAT MUG 1.25
INFLATABLE POLITICAL GLOBE      1.7
IVORY DINER WALL CLOCK  17.0
IVORY EMBROIDERED QUILT 71.5
IVORY ENCHANTED FOREST PLACEMAT 3.3
IVORY GIANT GARDEN THERMOMETER  17.85
IVORY KITCHEN SCALES    8.5
IVORY KNITTED MUG COSY  4.949999999999999
JAM JAR WITH GREEN LID  2.42
JAM JAR WITH PINK LID   1.5699999999999998
JAM MAKING SET PRINTED  15.949999999999996
JAM MAKING SET WITH JARS        29.25
JAZZ HEARTS PURSE NOTEBOOK      0.85
JOY WOODEN BLOCK LETTERS        4.95
JUMBO  BAG BAROQUE BLACK WHITE  7.5
JUMBO BAG CHARLIE AND LOLA TOYS 5.9
JUMBO BAG DOLLY GIRL DESIGN     5.85
JUMBO BAG OWLS  5.85
JUMBO BAG PINK POLKADOT 5.85
JUMBO BAG PINK VINTAGE PAISLEY  3.9
JUMBO BAG RED RETROSPOT 13.049999999999999
JUMBO BAG SCANDINAVIAN PAISLEY  1.95
JUMBO BAG STRAWBERRY    3.9
JUMBO BAG TOYS  1.95
JUMBO BAG WOODLAND ANIMALS      3.9
JUMBO SHOPPER VINTAGE RED PAISLEY       15.599999999999998
JUMBO STORAGE BAG SKULLS        3.9
JUMBO STORAGE BAG SUKI  7.8
KINGS CHOICE BISCUIT TIN        4.25
KINGS CHOICE GIANT TUBE MATCHES 5.1
KINGS CHOICE TEA CADDY  2.95
KITCHEN METAL SIGN      0.55
KNITTED UNION FLAG HOT WATER BOTTLE     20.34
LADIES & GENTLEMEN METAL SIGN   5.1
LADS ONLY TISSUE BOX    1.25
LANTERN CREAM GAZEBO    4.95
LARGE CHINESE STYLE SCISSOR     0.85
LARGE HANGING IVORY & RED WOOD BIRD     0.65
LARGE HEART MEASURING SPOONS    3.3
LARGE POPCORN HOLDER    4.75
LARGE ROUND WICKER PLATTER      5.95
LAVENDER INCENSE IN TIN 1.25
LETS GO SHOPPING COTTON TOTE BAG        2.25
LIGHT GARLAND BUTTERFILES PINK  3.37
LOLITA  DESIGN  COTTON TOTE BAG 2.25
```

Appendix 30: MapReduce Output

```
Administrator: Command Prompt
LOVE BUILDING BLOCK WORD        23.8
LOVEBIRD HANGING DECORATION WHITE       2.55
LUNCH BAG  BLACK SKULL. 4.949999999999999
LUNCH BAG CARS BLUE     3.3
LUNCH BAG DOLLY GIRL DESIGN     4.949999999999999
LUNCH BAG PINK POLKADOT 1.65
LUNCH BAG RED RETROSPOT 4.949999999999999
LUNCH BAG SPACEBOY DESIGN       4.949999999999999
LUNCH BAG SUKI  DESIGN  4.949999999999999
LUNCH BAG WOODLAND      1.65
LUNCH BOX I LOVE LONDON 3.9
LUNCH BOX WITH CUTLERY RETROSPOT        2.55
MAGIC DRAWING SLATE BAKE A CAKE 0.42
MAGIC DRAWING SLATE CIRCUS PARADE       1.26
MAGIC DRAWING SLATE DINOSAUR    0.42
MAGIC DRAWING SLATE DOLLY GIRL  0.84
MAGIC DRAWING SLATE GO TO THE FAIR      0.42
MAGIC DRAWING SLATE SPACEBOY    0.84
MAKE YOUR OWN FLOWERPOWER CARD KIT      5.9
MAKE YOUR OWN MONSOON CARD KIT  8.850000000000001
MAKE YOUR OWN PLAYTIME CARD KIT 2.95
MEASURING TAPE BABUSHKA BLUE    2.95
MEASURING TAPE BABUSHKA RED     5.9
METAL 4 HOOK HANGER FRENCH CHATEAU      11.8
METAL MERRY CHRISTMAS WREATH    1.95
METAL SIGN EMPIRE TEA   2.95
METAL SIGN HER DINNER IS SERVED 2.95
METAL SIGN TAKE IT OR LEAVE IT  2.95
MILK BOTTLE WITH GLASS STOPPER  6.95
MINI FUNKY DESIGN TAPES 0.85
MINI JIGSAW BAKE A CAKE 0.42
MINI JIGSAW CIRCUS PARADE       0.84
MINI JIGSAW DINOSAUR    0.42
MINI JIGSAW DOLLY GIRL  0.42
MINI JIGSAW SPACEBOY    0.84
MINI PAINT SET VINTAGE  1.9500000000000002
MINT KITCHEN SCALES     8.5
MIRRORED DISCO BALL     5.95
MIRRORED WALL ART FOXY  6.35
MIRRORED WALL ART GENTS 2.55
MULTI COLOUR SILVER T-LIGHT HOLDER      0.85
MULTICOLOUR  CONFETTI IN TUBE   1.65
NAMASTE SWAGAT INCENSE  0.24
NATURAL SLATE HEART CHALKBOARD  5.9
NATURAL SLATE RECTANGLE CHALKBOARD      1.65
NO JUNK MAIL METAL SIGN 1.25
NO SINGING METAL SIGN   2.1
NOEL WOODEN BLOCK LETTERS       5.95
OFFICE MUG WARMER BLACK+SILVER  2.95
```

Appendix 31: MapReduce Output

```
Administrator: Command Prompt
OFFICE MUG WARMER POLKADOT      2.95
ORANGE SCENTED SET/9 T-LIGHTS   2.55
ORGANISER WOOD ANTIQUE WHITE    8.5
OWL DOORSTOP    4.95
PACK 3 BOXES BIRD PANNETONE     5.85
PACK 3 BOXES CHRISTMAS PANNETONE        7.8
PACK 3 FIRE ENGINE/CAR PATCHES  1.25
PACK OF 12 BLUE PAISLEY TISSUES 0.29
PACK OF 12 CIRCUS PARADE TISSUES        0.29
PACK OF 12 HEARTS DESIGN TISSUES        0.58
PACK OF 12 LONDON TISSUES       0.8699999999999999
PACK OF 12 PINK PAISLEY TISSUES 0.29
PACK OF 12 PINK POLKADOT TISSUES        0.29
PACK OF 12 RED RETROSPOT TISSUES        1.16
PACK OF 12 SKULL TISSUES        0.29
PACK OF 12 SPACEBOY TISSUES     0.29
PACK OF 12 SUKI TISSUES 0.29
PACK OF 12 TRADITIONAL CRAYONS  0.42
PACK OF 12 WOODLAND TISSUES     0.29
PACK OF 20 NAPKINS PANTRY DESIGN        0.85
PACK OF 6 BIRDY GIFT TAGS       5.0
PACK OF 6 HANDBAG GIFT BOXES    7.6499999999999995
PACK OF 6 PANNETONE GIFT BOXES  7.6499999999999995
PACK OF 6 SWEETIE GIFT BOXES    5.1
PACK OF 60 DINOSAUR CAKE CASES  1.1
PACK OF 60 MUSHROOM CAKE CASES  0.55
PACK OF 60 PINK PAISLEY CAKE CASES      3.3
PACK OF 60 SPACEBOY CAKE CASES  0.55
PACK OF 72 RETROSPOT CAKE CASES 4.27
PACK OF 72 SKULL CAKE CASES     1.1
PAINT YOUR OWN CANVAS SET       1.65
PAINTED LIGHTBULB STAR+ MOON    0.42
PAINTED METAL HEART WITH HOLLY BELL     1.45
PAINTED METAL STAR WITH HOLLY BELLS     1.45
PAISLEY PATTERN  STICKERS       0.85
PANDA AND BUNNIES STICKER SHEET 0.85
PAPER BUNTING COLOURED LACE     2.95
PAPER BUNTING WHITE LACE        2.95
PAPER CHAIN KIT 50'S CHRISTMAS  22.4
PAPER CHAIN KIT LONDON  5.9
PAPER CHAIN KIT RETROSPOT       2.95
PAPER CHAIN KIT VINTAGE CHRISTMAS       22.4
PARTY CONE CHRISTMAS DECORATION 1.7
PARTY CONES CANDY ASSORTED      2.5
PARTY CONES CARNIVAL ASSORTED   1.25
PARTY INVITES FOOTBALL  0.85
PARTY INVITES JAZZ HEARTS       0.85
PARTY INVITES SPACEMAN  0.85
PARTY TIME PENCIL ERASERS       0.21
```

Appendix 32: MapReduce Output

```
Administrator: Command Prompt
PEACE WOODEN BLOCK LETTERS      6.95
PENCIL CASE LIFE IS BEAUTIFUL   2.95
PENS ASSORTED FUNNY FACE        0.85
PHOTO CLIP LINE 2.5
PHOTO CUBE      3.13
PHOTO FRAME CORNICE     5.9
PICNIC BASKET WICKER LARGE      9.95
PICNIC BASKET WICKER SMALL      11.9
PICTURE DOMINOES        2.9
PIGGY BANK RETROSPOT    7.6499999999999995
PINK  POLKADOT PLATE    1.69
PINK & WHITE BREAKFAST TRAY     5.95
PINK 3 PIECE POLKADOT CUTLERY SET       3.75
PINK B'FLY C/COVER W BOBBLES    5.95
PINK BREAKFAST CUP AND SAUCER   2.95
PINK CREAM FELT CRAFT TRINKET BOX       2.5
PINK DOUGHNUT TRINKET POT       1.65
PINK DRAWER KNOB ACRYLIC EDWARDIAN      1.25
PINK FLOWER FABRIC PONY 3.75
PINK HEART DOTS HOT WATER BOTTLE        7.5
PINK HEART SHAPE EGG FRYING PAN 1.25
PINK HEARTS PAPER GARLAND       3.3
PINK NEW BAROQUECANDLESTICK CANDLE      5.9
PINK OVAL JEWELLED MIRROR       5.95
PINK PAISLEY SQUARE TISSUE BOX  1.25
PINK PARTY BAGS 2.1
PINK ROSE FABRIC MIRROR 1.25
PINK SWEETHEART BRACELET        4.25
PINK UNION JACK  LUGGAGE TAG    2.5
PIZZA PLATE IN BOX      7.5
PLACE SETTING WHITE HEART       1.26
PLACE SETTING WHITE STAR        0.84
PLASTERS IN TIN CIRCUS PARADE   1.65
PLASTERS IN TIN SKULLS  6.6
PLASTERS IN TIN SPACEBOY        8.25
PLASTERS IN TIN STRONGMAN       3.3
PLASTERS IN TIN VINTAGE PAISLEY 9.9
PLASTERS IN TIN WOODLAND ANIMALS        4.949999999999999
POLKA DOT RAFFIA FOOD COVER     3.75
POLKADOT MUG PINK       1.65
POLKADOT PEN    0.85
POLKADOT RAIN HAT       0.74
POLYESTER FILLER PAD 40x40cm    1.45
POLYESTER FILLER PAD 45x30cm    1.45
POLYESTER FILLER PAD 45x45cm    1.55
POPPY'S PLAYHOUSE BEDROOM       2.1
POPPY'S PLAYHOUSE KITCHEN       2.1
POSTAGE 51.0
POTTING SHED SEED ENVELOPES     1.25
```

Appendix 33: MapReduce Output

```
Administrator: Command Prompt
POTTING SHED SOW 'N' GROW SET   4.25
PURPLE DRAWERKNOB ACRYLIC EDWARDIAN     1.25
PURPLE SWEETHEART BRACELET      4.25
RAINY LADIES BIRTHDAY CARD      0.42
RECIPE BOX BLUE SKETCHBOOK DESIGN       2.95
RECIPE BOX PANTRY YELLOW DESIGN 2.95
RECIPE BOX RETROSPOT    2.95
RECIPE BOX WITH METAL HEART     7.95
RECYCLING BAG RETROSPOT 6.300000000000001
RED  HARMONICA IN BOX   3.75
RED 3 PIECE RETROSPOT CUTLERY SET       7.5
RED CHARLIE+LOLA PERSONAL DOORSIGN      0.38
RED COAT RACK PARIS FASHION     4.95
RED DINER WALL CLOCK    17.0
RED DRAWER KNOB ACRYLIC EDWARDIAN       1.25
RED ENCHANTED FOREST PLACEMAT   1.65
RED FLOWER CROCHET FOOD COVER   3.75
RED GINGHAM ROSE JEWELLERY BOX  1.65
RED GLASS TASSLE BAG CHARM      2.95
RED HANGING HEART T-LIGHT HOLDER        8.45
RED HEART LUGGAGE TAG   1.25
RED KITCHEN SCALES      17.0
RED METAL BEACH SPADE   1.25
RED RETROSPOT CAKE STAND        21.9
RED RETROSPOT MINI CASES        15.9
RED RETROSPOT MUG       2.95
RED RETROSPOT OVEN GLOVE        1.06
RED RETROSPOT OVEN GLOVE DOUBLE 5.9
RED RETROSPOT PLATE     1.69
RED RETROSPOT TAPE      0.65
RED RETROSPOT TISSUE BOX        1.25
RED RETROSPOT UMBRELLA  11.9
RED RETROSPOT WRAP      0.84
RED SHARK HELICOPTER    7.95
RED STAR CARD HOLDER    2.95
RED TEA TOWEL CLASSIC DESIGN    2.5
RED TOADSTOOL LED NIGHT LIGHT   7.85
RED WHITE SCARF  HOT WATER BOTTLE       3.95
RED WOOLLY HOTTIE WHITE HEART.  31.590000000000003
REGENCY CAKESTAND 3 TIER        87.45
RETRO COFFEE MUGS ASSORTED      5.49
RETRO LONGBOARD IRONING BOARD COVER     3.75
RETROSPOT BABUSHKA DOORSTOP     3.75
RETROSPOT CHILDRENS APRON       7.8
RETROSPOT CIGAR BOX MATCHES     5.0
RETROSPOT GIANT TUBE MATCHES    2.55
RETROSPOT HEART HOT WATER BOTTLE        33.95
RETROSPOT LAMP  9.95
RETROSPOT LARGE MILK JUG        9.2
```

Appendix 34: MapReduce Output

Administrator: Command Prompt

```
RETROSPOT PARTY BAG + STICKER SET       3.3
RETROSPOT RED WASHING UP GLOVES 4.2
RETROSPOT SMALL TUBE MATCHES    3.3
RETROSPOT TEA SET CERAMIC 11 PC 4.25
REX CASH+CARRY JUMBO SHOPPER    2.8499999999999996
RIBBON REEL CHRISTMAS PRESENT   1.65
RIBBON REEL CHRISTMAS SOCK BAUBLE       3.3
RIBBON REEL MAKING SNOWMEN      3.3
RIBBON REEL SNOWY VILLAGE       3.3
RIBBON REEL SOCKS AND MITTENS   1.65
RIBBON REEL STRIPES DESIGN      4.949999999999999
RIDGED GLASS FINGER BOWL        1.45
RIDGED GLASS STORAGE JAR CREAM LID      11.25
ROCKING HORSE RED CHRISTMAS     0.72
ROMANTIC IMAGES NOTEBOOK SET    4.25
ROMANTIC PINKS RIBBONS  1.25
ROSE CARAVAN DOORSTOP   6.75
ROSE COTTAGE KEEPSAKE BOX       38.349999999999994
ROSES REGENCY TEACUP AND SAUCER 5.9
ROTATING LEAVES T-LIGHT HOLDER  1.25
ROTATING SILVER ANGELS T-LIGHT HLDR     15.3
ROUND SNACK BOXES SET OF 4 FRUITS       2.95
ROUND SNACK BOXES SET OF 4 SKULLS       5.9
ROUND SNACK BOXES SET OF4 WOODLAND      11.8
RUSTIC  SEVENTEEN DRAWER SIDEBOARD      165.0
S/15 SILVER GLASS BAUBLES IN BAG        1.25
S/4 VALENTINE DECOUPAGE HEART BOX       5.9
S/6 SEW ON CROCHET FLOWERS      1.25
S/6 WOODEN SKITTLES IN COTTON BAG       8.850000000000001
SANDWICH BATH SPONGE    2.5
SAVE THE PLANET COTTON TOTE BAG 2.25
SAVE THE PLANET MUG     4.24
SCANDINAVIAN REDS RIBBONS       7.5
SCOTTIE DOG HOT WATER BOTTLE    34.65
SET 10 LIGHTS NIGHT OWL 6.75
SET 12 KIDS  WHITE CHALK STICKS 0.42
SET 12 LAVENDER  BOTANICAL T-LIGHTS     2.95
SET 12 RETRO WHITE CHALK STICKS 0.84
SET 2 TEA TOWELS I LOVE LONDON  29.499999999999996
SET 20 NAPKINS FAIRY CAKES DESIGN       0.85
SET 3 WICKER OVAL BASKETS W LIDS        56.849999999999994
SET 6 FOOTBALL CELEBRATION CANDLES      1.25
SET 7 BABUSHKA NESTING BOXES    55.25
SET OF 16 VINTAGE PISTACHIO CUTLERY     12.75
SET OF 2 CHRISTMAS DECOUPAGE CANDLE     1.25
SET OF 2 TEA TOWELS APPLE AND PEARS     8.850000000000001
SET OF 2 TINS JARDIN DE PROVENCE        4.95
SET OF 2 WOODEN MARKET CRATES   12.75
SET OF 20 KIDS COOKIE CUTTERS   4.2
```

Appendix 35: MapReduce Output

Administrator: Command Prompt

```
SET OF 20 VINTAGE CHRISTMAS NAPKINS     1.7
SET OF 3 BLACK FLYING DUCKS     4.65
SET OF 3 BUTTERFLY COOKIE CUTTERS       1.25
SET OF 3 COLOURED  FLYING DUCKS 10.100000000000001
SET OF 3 GOLD FLYING DUCKS      6.35
SET OF 3 HEART COOKIE CUTTERS   2.5
SET OF 3 NOTEBOOKS IN PARCEL    4.949999999999999
SET OF 4 ENGLISH ROSE COASTERS  1.25
SET OF 4 ENGLISH ROSE PLACEMATS 3.75
SET OF 4 NAPKIN CHARMS CUTLERY  2.55
SET OF 4 NAPKIN CHARMS LEAVES   2.55
SET OF 4 NAPKIN CHARMS STARS    2.55
SET OF 6 FUNKY BEAKERS  2.95
SET OF 6 GIRLS CELEBRATION CANDLES      1.25
SET OF 6 KASHMIR FOLKART BAUBLES        4.25
SET OF 6 SOLDIER SKITTLES       22.5
SET OF 6 T-LIGHTS SANTA 8.850000000000001
SET OF 6 T-LIGHTS SNOWMEN       5.9
SET OF 6 T-LIGHTS TOADSTOOLS    8.850000000000001
SET OF 72 GREEN PAPER DOILIES   4.35
SET OF 72 RETROSPOT PAPER  DOILIES      1.45
SET OF 9 HEART SHAPED BALLOONS  1.25
SET/10 BLUE POLKADOT PARTY CANDLES      3.75
SET/10 IVORY POLKADOT PARTY CANDLES     1.25
SET/10 PINK POLKADOT PARTY CANDLES      6.25
SET/10 RED POLKADOT PARTY CANDLES       5.0
SET/12 TAPER CANDLES    0.85
SET/2 RED RETROSPOT TEA TOWELS  5.9
SET/20 RED RETROSPOT PAPER NAPKINS      1.49
SET/4 MODERN VINTAGE COTTON NAPKINS     2.95
SET/5 RED RETROSPOT LID GLASS BOWLS     8.45
SET/6 RED SPOTTY PAPER CUPS     0.65
SET/6 RED SPOTTY PAPER PLATES   1.49
SET/9 CHRISTMAS T-LIGHTS SCENTED        2.5
SILK PURSE BABUSHKA PINK        10.05
SILK PURSE BABUSHKA RED 6.7
SILVER  CANDLEPOT JARDIN        3.75
SILVER GLASS T-LIGHT SET        2.95
SILVER HANGING T-LIGHT HOLDER   4.949999999999999
SILVER LOOKING MIRROR   4.95
SILVER PLATE CANDLE BOWL SMALL  2.95
SINGLE HEART ZINC T-LIGHT HOLDER        1.9
SKULL SHOULDER BAG      1.65
SKULLS  WATER TRANSFER TATTOOS  0.85
SKULLS AND CROSSBONES WRAP      0.42
SKULLS SQUARE TISSUE BOX        2.5
SKULLS STORAGE BOX SMALL        2.1
SKULLS WRITING SET      1.65
SLATE TILE NATURAL HANGING      1.65
```

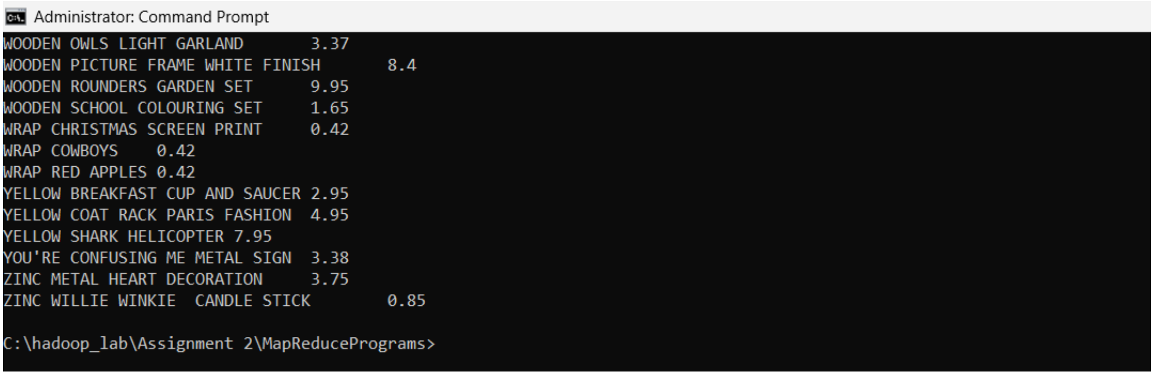
Appendix 36: MapReduce Output

```
Administrator: Command Prompt
SLEEPING CAT ERASERS    0.21
SMALL CHINESE STYLE SCISSOR     0.42
SMALL GLASS HEART TRINKET POT   3.95
SMALL HANGING IVORY/RED WOOD BIRD       0.42
SMALL HEART FLOWERS HOOK        0.85
SMALL HEART MEASURING SPOONS    0.85
SMALL LICORICE DES PINK BOWL    1.25
SMALL POPCORN HOLDER    6.669999999999999
SMALL RED BABUSHKA NOTEBOOK     0.85
SMALL REGAL  SILVER CANDLEPOT   2.95
SMALL STRIPES CHOCOLATE GIFT BAG        0.85
SNOWFLAKE PORTABLE TABLE LIGHT  2.95
SOLDIERS EGG CUP        2.5
SPACEBOY CHILDRENS EGG CUP      1.25
SPACEBOY LUNCH BOX      5.85
SQUARECUSHION COVER PINK UNION FLAG     13.5
STAR  T-LIGHT HOLDER    2.9
STAR DECORATION PAINTED ZINC    1.3
STAR PORTABLE TABLE LIGHT       8.850000000000001
STAR WOODEN CHRISTMAS DECORATION        1.7
STARS GIFT TAPE 0.65
STRAWBERRY BATH SPONGE  1.25
STRAWBERRY CERAMIC TRINKET BOX  1.25
STRAWBERRY CHARLOTTE BAG        1.7
STRAWBERRY LUNCH BOX WITH CUTLERY       2.55
STRAWBERRY RAFFIA FOOD COVER    3.75
STRAWBERRY SHOPPER BAG  1.25
STRING OF STARS CARD HOLDER     2.95
SUKI  SHOULDER BAG      1.65
SWALLOW SQUARE TISSUE BOX       2.5
SWEETHEART CAKESTAND 3 TIER     9.95
SWEETHEART WIRE WALL TIDY       9.95
SWIRLY CIRCULAR RUBBERS IN BAG  0.42
SWISS ROLL TOWEL- CHOCOLATE  SPOTS      5.9
T-LIGHT GLASS FLUTED ANTIQUE    1.25
T-LIGHT HOLDER HANGING LACE     1.25
TEA TIME DES TEA COSY   2.55
TEA TIME TABLE CLOTH    10.65
THREE CANVAS LUGGAGE TAGS       1.95
TOADSTOOL MONEY BOX     2.95
TOAST ITS - HAPPY BIRTHDAY      1.25
TOILET METAL SIGN       0.55
TOMATO CHARLIE+LOLA COASTER SET 2.95
TOOL BOX SOFT TOY       8.95
TOTE BAG I LOVE LONDON  4.2
TOY TIDY DOLLY GIRL DESIGN      2.1
TOY TIDY PINK POLKADOT  1.85
TRADITIONAL CHRISTMAS RIBBONS   2.5
TRADITIONAL WOODEN CATCH CUP GAME       1.25
```

Appendix 37: MapReduce Output

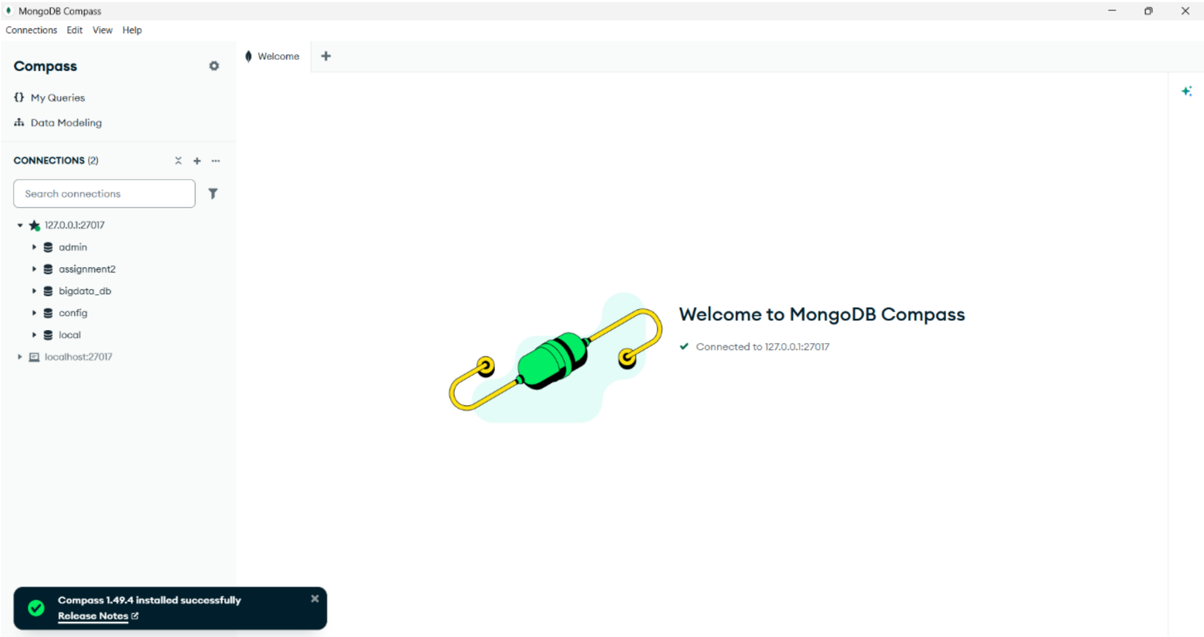
```
Administrator: Command Prompt
TRADITIONAL WOODEN SKIPPING ROPE        5.0
TRAVEL SEWING KIT       3.3
TRAY- BREAKFAST IN BED  12.75
TRIPLE PHOTO FRAME CORNICE      19.9
TURQUOISE CHRISTMAS TREE        1.25
TV DINNER TRAY AIR HOSTESS      4.95
TV DINNER TRAY DOLLY GIRL       4.95
TV DINNER TRAY VINTAGE PAISLEY  4.95
UNION JACK FLAG LUGGAGE TAG     2.5
UNION STRIPE WITH FRINGE  HAMMOCK       7.95
URBAN BLACK RIBBONS     1.25
VANILLA SCENT CANDLE JEWELLED BOX       4.25
VICTORIAN  METAL POSTCARD SPRING        1.69
VICTORIAN GLASS HANGING T-LIGHT 1.25
VICTORIAN SEWING BOX LARGE      36.45
VICTORIAN SEWING BOX MEDIUM     7.95
VICTORIAN SEWING BOX SMALL      11.9
VINTAGE BILLBOARD DRINK ME MUG  5.49
VINTAGE BILLBOARD LOVE/HATE MUG 4.24
VINTAGE GLASS COFFEE CADDY      5.95
VINTAGE HEADS AND TAILS CARD GAME       5.0
VINTAGE KEEPSAKE BOX PARIS DAYS 6.35
VINTAGE PAISLEY STATIONERY SET  8.850000000000001
VINTAGE SEASIDE JIGSAW PUZZLES  3.75
VINTAGE SNAKES & LADDERS        3.75
VINTAGE SNAP CARDS      0.85
VINTAGE UNION JACK APRON        13.9
VINTAGE UNION JACK BUNTING      8.5
VINTAGE UNION JACK CUSHION COVER        9.9
WASHROOM METAL SIGN     2.9
WELCOME  WOODEN BLOCK LETTERS   9.95
WHITE BELL HONEYCOMB PAPER      1.65
WHITE CHRISTMAS GARLAND STARS TREES     3.75
WHITE HANGING HEART T-LIGHT HOLDER      29.65
WHITE LOVEBIRD LANTERN  2.95
WHITE METAL LANTERN     16.95
WHITE SKULL HOT WATER BOTTLE    15.0
WHITE SPOT RED CERAMIC DRAWER KNOB      2.5
WHITE WIRE EGG HOLDER   4.95
WHITE WOOD GARDEN PLANT LADDER  9.95
WICKER STAR     4.2
WICKER WREATH SMALL     1.45
WOOD 2 DRAWER CABINET WHITE FINISH      25.75
WOOD BLACK BOARD ANT WHITE FINISH       6.45
WOOD S/3 CABINET ANT WHITE FINISH       35.75
WOODEN ADVENT CALENDAR RED      16.95
WOODEN BOX OF DOMINOES  3.75
WOODEN FRAME ANTIQUE WHITE      13.149999999999999
WOODEN HEART CHRISTMAS SCANDINAVIAN     0.85
```

Appendix 38: MapReduce Output

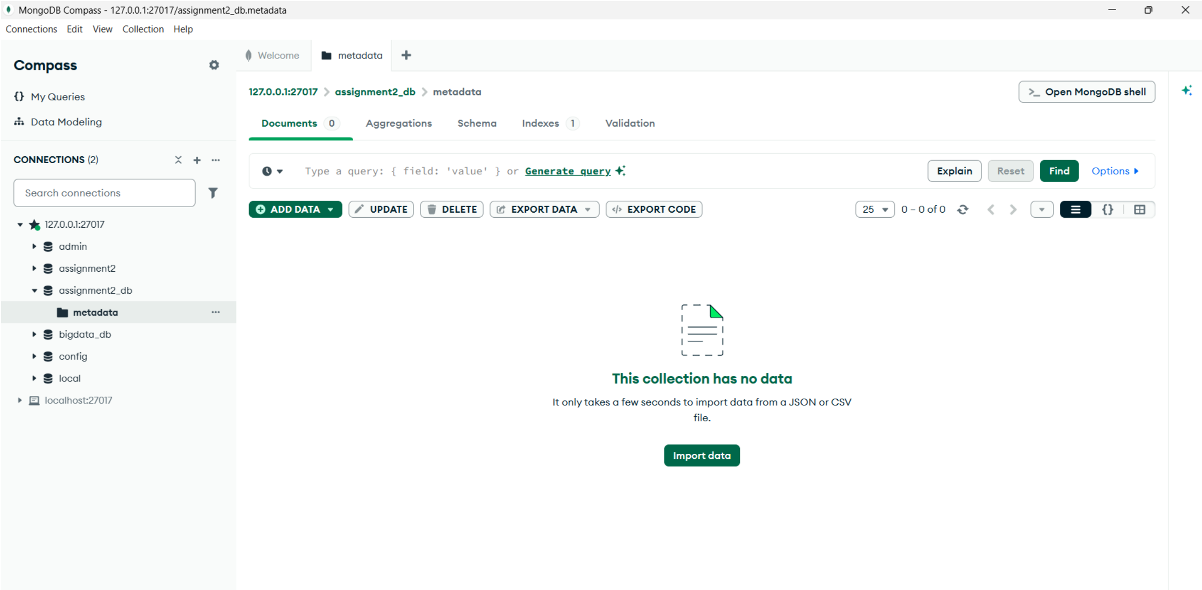


Appendix 39: MapReduce Output

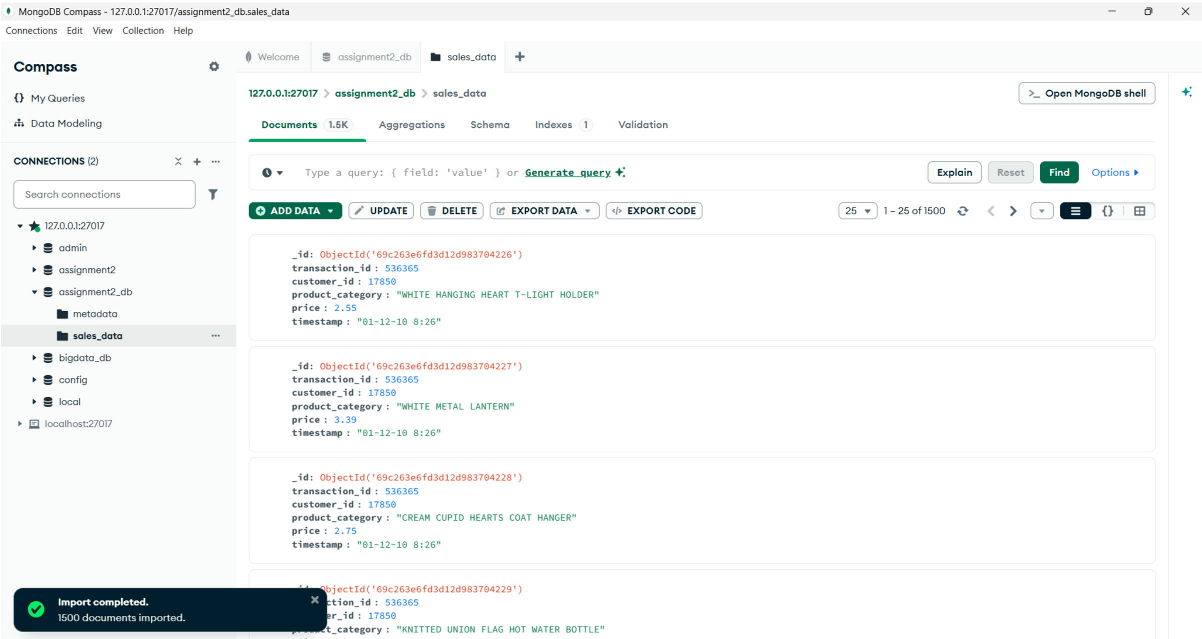
Part 7:- MongoDB Integration



Appendix 40: MongoDB Compass Connection

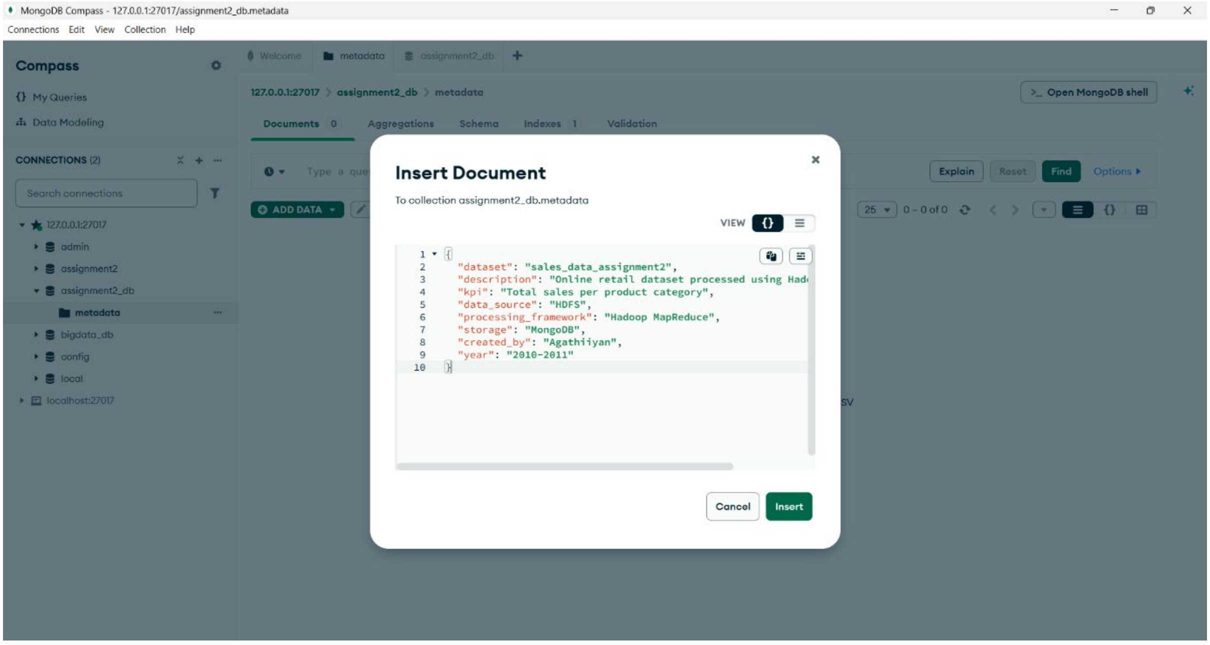


Appendix 41: MongoDB Database Structure

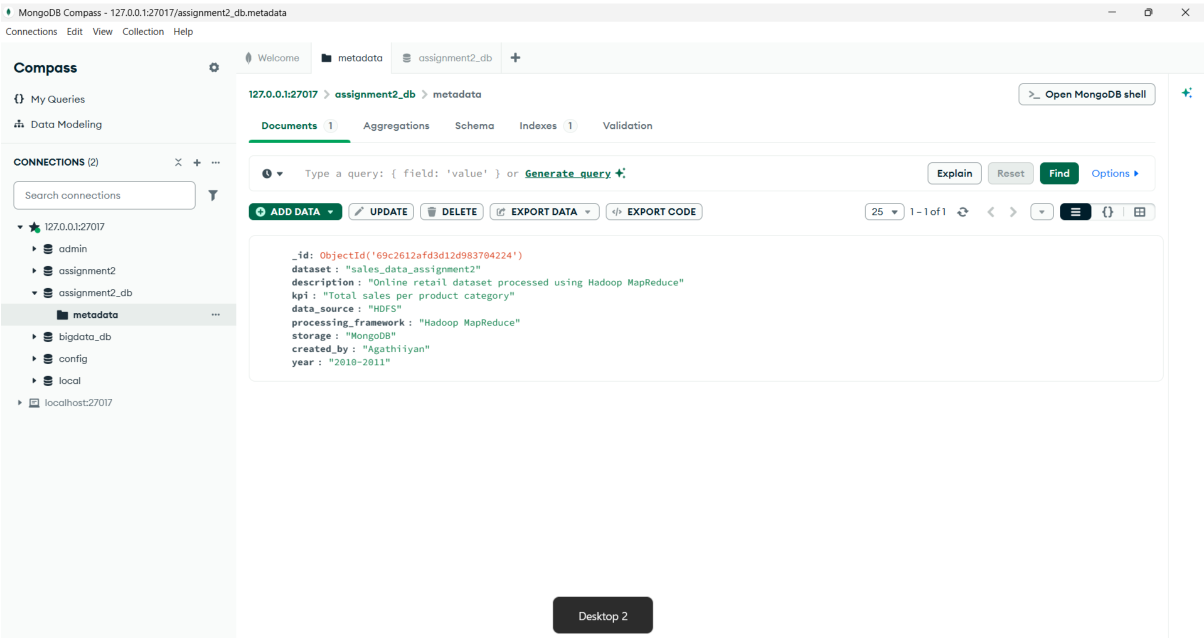


Appendix 42: Data Imported into MongoDB Collection

Part 8:- Metadata



Appendix 43: Metadata Structure in JSON Format



Appendix 44: Metadata Inserted into MongoDB